GPW WATS 3.01

Market Data Protocol

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1. correcting errors in the documentation or in the software;
2. clarification of the documentation content or removing ambiguity;
3. implementation of approved change requests or;
4. regulatory changes.

2. Preface

This document has been prepared by Warsaw Stock Exchange in order to help in the implementation process of GPW WATS trading platform".

2.1. Target Audience

This document has been prepared to development staff, Independent Software Vendors who produce software integrated with GPW WATS, analysts, market participants and all clients who want to deepen their knowledge about GPW WATS.

2.2. Document's Purpose

The purpose of this document is to provide a full description of Market Data Gateway, which is part of GPW WATS.

This document contains a description of trading port behavior, message types and their fields.

2.3. Associated Documents

GPW WATS 3.01 Market Data Protocol is a part of GPW WATS documentation set.

Please check the following documents to learn about the construction of Trading System.

- GPW WATS 1.01 Trading System

Please check the documentation of the trading protocols supported by GPW WATS.

- GPW WATS 2.01 Native Order Gateway Specification
- GPW WATS 2.02 FIX Order Gateway Specification .

Please check the description of the communication with Data Distribution Service.

- GPW WATS 3.01 Market Data Protocol (this document)

Please check the additional documentation which explains other services provided within GPW WATS.

- GPW WATS 4.01 Drop Copy Gateway
- GPW WATS 4.02 Post Trade Gateway
- GPW WATS 5.01 Risk Management Gateway.

Please check the additional documentation describing the following:

- GPW WATS 2.03 Rejection Codes
- GPW WATS 2.04 BenDec Message Definition Format
- GPW WATS 6.01 Connectivity

It is recommended that you read the GPW WATS 1.01 Trading System document first.

3. Document History

Version	Date	Description
0.51	29.06.2023	The initial publication of the documentation.

0.52	26.07.2023	Messages updated in protocol v0.52:
		• EndOfSnapshot
		  o New field lastBBOSeqNum
		• Instrument
		  o New field referencePrice
		• ReplayRequest

0.53	16.08.2023	Messages updated in protocol v0.53: • EndOfSnapshot o Field lastBBOSeqNum removed
0.54	19.09.2023	Messages updated in protocol v0.54: • TradingPhaseScheduleEntry • Deleted attributes: o referencePriceLogicId

0.55	17.10.2023	Section 5.4.2: Updated sequence of messages in the Snapshot service.
		Messages updated in protocol v0.55
		Instrument
		• status – new dictionary values:
		  o Active
		  o Inactive
		  o MarketOperationsSuspension
		  o OutsideCollarsStatic
		  o OutsideCollarsDynamic
		  o RegulatorySuspension
		  o TechnicalHalt
		• issuerRegCountry added values:
		  o PRT
		  o SVK
		InstrumentStatusChange
		• status – added values:
		  o OutsideCollarsStatic
		  o OutsideCollarsDynamic
		MarketStructure
		• Deleted attributes
		  o code
		  o structureType
		• New attributes:
		• marketModelType, with values
		  o CLOB
		  o BLOCK
		  o HYBRID
		  o CROSS
		  o NotApplicable
		SessionSummary
		• status – added values:
		  o OutsideCollarsStatic
		  o OutsideCollarsDynamic

0.56	17.11.2023	Document Changes
		• OrderBookEvent supersedes the functionality of the ClearBook message. This change specifically affects the order of messages after auction uncrossing.
		Message examples were updated throughout the document to reflect the latest version of the Market Data protocol. The affected messages are:
		• Trading:
		  o OrderAdd
		  o OrderExecute
		  o OrderModify

Version	Date	Description
		o OrderDelete • Status: o InstrumentStatusChange o TradeCollars o OrderCollars • Reference data: o MarketStructure o Instrument o TickTableEntry o TradingPhaseScheduleEntry o CalendarException o WeekPlan o CollarTableEntry o CollarGroup • Session management: o StartOfTechnicalSession o ReferenceDataStart o ReferenceDataEnd Deleted messages: • ClearBook • TestMax Added messages: • OrderBookEvent - Order book event for given financial instrument • TestEvent – a test event TestEvent • scenarioName – name of test scenario • eventType – type of test event: o Flush - Flush the test scenario o ScenarioStart - Start of test scenario o ScenarioEnd – End of test scenario OrderBookEvent • instrumentID • eventType: - type of book event o Clear – clear order book o ResendStart - Order book resend start (resend will begin shortly) o ResendEnd - Order book resend end (resend is done) Messages updated in protocol v0.56 Instrument • New attribute: • initialPhaseId - Id of starting phase for the instrument OrderDelete • New attribute: • instrumentID – ID of financial instrument OrderModify • New attribute: • instrumentID – ID of financial instrument
0.57	30.11.2023	Document Changes • Section 6.2 was updated to clarify the usage of the orderId and publicOrderId identifiers in Market Data. New messages: OpenPositions: The number of open positions in a financial instrument. • Description of the message and its purpose added to Section 6.4 Messages updated in protocol v0.57 Trade • New attribute: o settlementValue o settlementDate TradingPhaseScheduleEntry • Updated attribute:

Version	Date	Description

		• tradingPhaseType – added values:
		  ○ Hybrid – Hybrid normal trading
		  ○ HybridPause – Hybrid pause
		• New attribute:
		• hybridPauseId - ID of Hybrid pause phase.

0.58	18.12.2023	Document Changes
		• Section 5.3 updated to reflect changes to structure of BBO data, now includes OpenPositions message.
		• Section 6.3.1 PriceLevelSnapshot, provides a more precise description of the behavior of the message in the BBO during auctions.
		• Section 6.3.2 describes the sequence of messages during an auction in the BBO stream.
		Messages updated in protocol v0.58
		AuctionSummary
		• Updated attribute:
		• auctionType – added values:
		  ○ UnsuspensionAuction
		Instrument
		• Updated attribute:
		• status – added values:
		  ○ HybridNoQuotes - Hybrid no valid quotes.
		  ○ HybridPause - Hybrid pause.
		InstrumentStatusChange
		• Updated attribute:
		• status – added values:
		  ○ HybridNoQuotes - Hybrid no valid quotes.
		  ○ HybridPause - Hybrid pause.
		InstrumentSummary
		• Updated attribute:
		• adjustedClosingPriceReason – removed values:
		  ○ TickTableChange
		  ○ MultipleCA
		SessionSummary
		• Updated attribute:
		• status – added values:
		  ○ HybridNoQuotes - Hybrid no valid quotes.
		  ○ HybridPause - Hybrid pause.
		TradingPhaseScheduleEntry
		• Updated attribute:
		• tradingPhaseType –updated values:
		  ○ HybridPause updated to HybridPreTrade
		  ○ added UnsuspensionAuction - Auction after suspension.
		• auctionType – added values:
		  ○ UnsuspensionAuction - Auction after suspension.
		• Deleted attribute:
		  ○ hybridPauseId

0.59	20.01.2024	Messages updated in protocol v0.59 Instrument • nominalValue, type changed to Price from Number • multiplier, type changed to Multiplier from Number
0.62	25.03.2024	Document Changes • Test, TestEvent and Text messages are deprecated • News takes the place of Text message • New description for TopPriceLevelUpdate to emphasize difference from PriceLevelSnapshot • OrderBookEvent supersedes the functionality of the ClearBook message. This change specifically affects the order of messages after auction uncrossing.

<table>
<tr><th>Version</th><th>Date</th><th>Description</th></tr>
<tr><td></td><td></td><td>Message examples were updated throughout the document to reflect the latest version of the Market Data protocol. The affected messages are:

Trading:
OrderAdd
OrderExecute
OrderModify
OrderDelete
Auction:
OrderBookEvent
AuctionUpdate
AuctionSummary
Status:
InstrumentStatusChange
TradeCollars
OrderCollars
Trade
Reference data:
MarketStructure
Instrument
TickTableEntry
TradingPhaseScheduleEntry
CalendarException
WeekPlan
CollarTableEntry
CollarGroup
AccruedInterestTableEntry
Session management:
StartOfTechnicalSession
ReferenceDataStart
ReferenceDataEnd

Messages updated in protocol v0.60
Deleted Messages:
Test
TestEvent
Text
CalendarException
Removed attribute:
calendarExceptionRecurrent
Updated attribute:
calendarExceptionType, now contains values:
Closed The day is closed for trading.
Open The day is open for trading.
IndexParams
New attribute:
dateValidity Date of validity of index.
Instrument
Updated attribute:
marketModelType: reordered dictionary of values, new value IPO
MarketStructure
Updated attribute:
marketModelType: Market model type, reordered dictionary with added value:
IPO: Initial public offering
TradingPhaseScheduleEntry
Removed attribute:
tradingPhaseStopTime
auctionUncrossing
</td></tr>
</table>

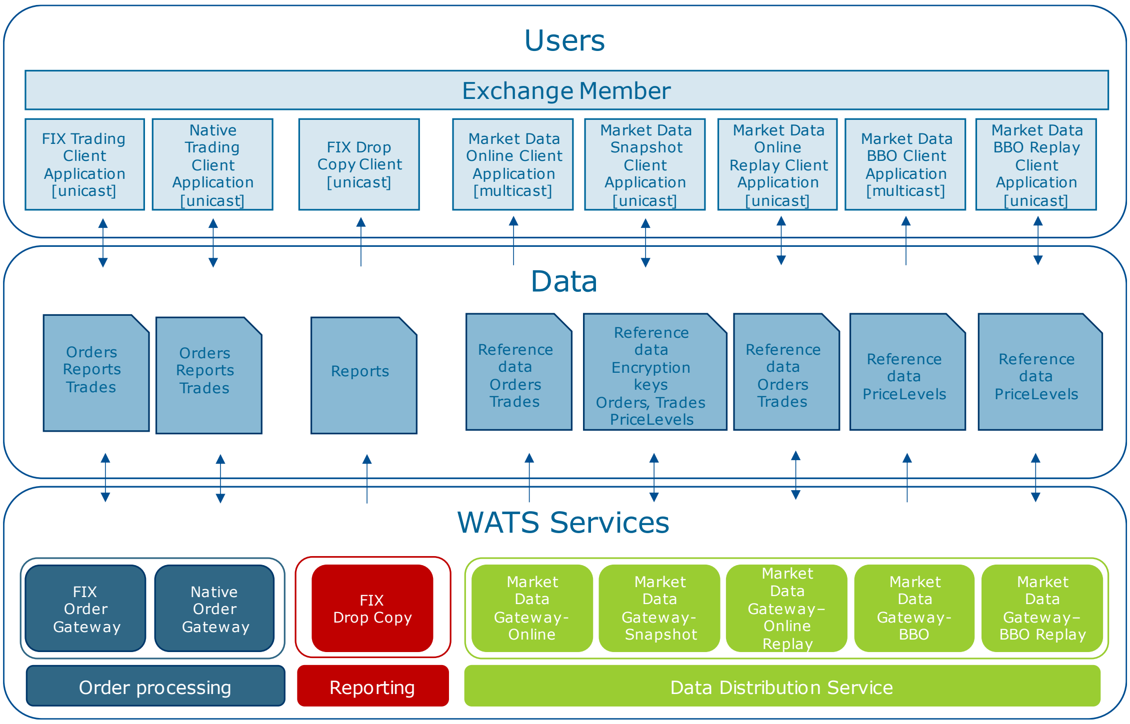
Version	Date	Description

		Updated attribute:
		• tradingPhaseDetailedType: reordered dictionary with new values,
		  ○ HybridBuyOnly: Hybrid buy only trading
		  ○ Ipo: Initial public offering phase
		New attribute:
		• uncrossing: True if phase includes an uncrossing.
		• lastAuctionPhaseId: ID of last auction phase.

4. GPW WATS MARKET DATA

GPW WATS Market Data is a Native protocol application serviced by the Data Distribution Service (DDS) and provides a range of market data services to users. The DDS is a complex framework designed to provide maximum flexibility in data management to the Market and its users. The scope of the Data Distribution System is highlighted in green in the following diagram.

Figure 1 - GPW WATS Services



GPW WATS Market Data provides users with full-depth and best bid-offer datasets. The data is provided in real time broadcasts through three separate channels, Snapshot, Replay and Stream:

- Snapshot provides reference data and encryption keys for decrypting information sent through Stream and Replay. It can be accessed at any time during the trading day to obtain the full-depth order book and a complete record of execution reports and enriched trades.
- Stream is the principal market data channel for disseminating real time order and transaction data over multicast. It publishes among others the full depth order book, BBO levels, as well as auction related statistics.
- Replay is a mechanism for recovering information lost by Stream during multicast transmission.

DDS Market Data can be configured in various ways depending on the needs of users and Markets. Different DDS connections can be configured to carry separate streams of information. For example, the Full Depth broadcast can define several connections each conveying full depth data related to specific Markets. Similarly, for BBO data feeds.

4.1. DATASETS

GPW WATS offers two views of the market: Full Depth and Best Bid-Offer (BBO). The Full Depth dataset provides full information on the current state of the order book including market depth, while the Best Bid-Offer provides up to 10 levels of the Order Book.

The table below lists the scope of the data distributed in each dataset.

	Real-time data	
Category	**Full Depth**	**Best Bid Offer**
Reference Data messages	Reference data: • market structure • instruments • trading phases • tick table entry • calendar exceptions • collar table entry • collar logic • reference prices • trading week plan	Reference data: • market structure • instruments • trading phases • tick table entry • calendar exceptions • collar table entry • collar logic • reference prices • trading week plan
Trading messages	Full order book state: • order additions/ modifications/ cancellations • trades • order and trade collar values	BBO information: • 10 BBO price levels • Trades • order and trade collar values
Auction messages	Auctions: • signal to clear book for an upcoming auction • auction type • auction price/quantity • indicative matching price / indicative matching volume • buy/sell quantity • BBO price/quantity if uncrossed book	Auctions: • auction type • auction price/quantity • indicative matching price / indicative matching volume • buy/sell quantity • BBO price/quantity if uncrossed book
Status messages	Changes in trading phases	Changes in trading phases

4.1.1. FULL DEPTH DATA

During a continuous trading phase (Continuous Price and Time) and the trade at last (Continuous Trading at Closing Price), Full Depth Market Data broadcasts all order messages received by the Market and can be used by a participant to reconstruct the state of the Order Book at any point in time.

During Auctions, Full Depth only disseminates auction related statistics such as the Indicative Matching Price and Indicative Matching Volume, or the best buy and sell levels. At the end of each auction (the uncrossing), Full Depth sends auction summary information. During auctions, unlike in continuous time trading, no full depth order book data is published.

In continuous time trading and continuous trading at fixed price (trade-at-last) Full Depth disseminates:

- OrderAdd, OrderModify, OrderDelete messages disseminated by the exchange,
- OrderExecute messages signaling that an order was filled,
- Trade messages which contain enriched trade information.

During Auctions Full Depth only conveys Auction related information:

- During auctions, the AuctionUpdate message continuously updates market participants on the state of the auction by publishing IMP/IMV or the best buy and sell levels
- At Uncrossing the AuctionSummary message provides a full auction summary including price and volume

4.1.2. BEST BID OFFER

In addition to full depth data, GPW WATS provides BBO data to users. By default, up to 10 levels are provided simultaneously. The key features of the BBO dataset are:

- 10 BBO levels for each instrument traded on the platform. Users can subscribe for up to 10 BBO levels of data if they do not need the Full Depth dataset.

4.2. SERVICE LEVELS

Market participants have the option to subscribe to either the Full Depth, the BBO dataset or both. Access rights will be managed by means of encryption. Through the Snapshot service, users will receive encryption keys enabling them to decrypt the relevant Market Data content which corresponds to Full Depth and BBO data. In addition, the content of the Snapshot will reflect each user's elections and will be customized to relay BBO and Full Depth content.

An example of service-level configuration is the option for a user to receive Trade and OrderCollars/TradeCollars messages in their BBO Stream. By default, these messages are not included in the BBO Stream, but can be activated with the appropriate subscription.

4.3. CHANNELS: SNAPSHOT, REPLAY AND STREAM.

4.3.1. GENERAL INFORMATION FLOW

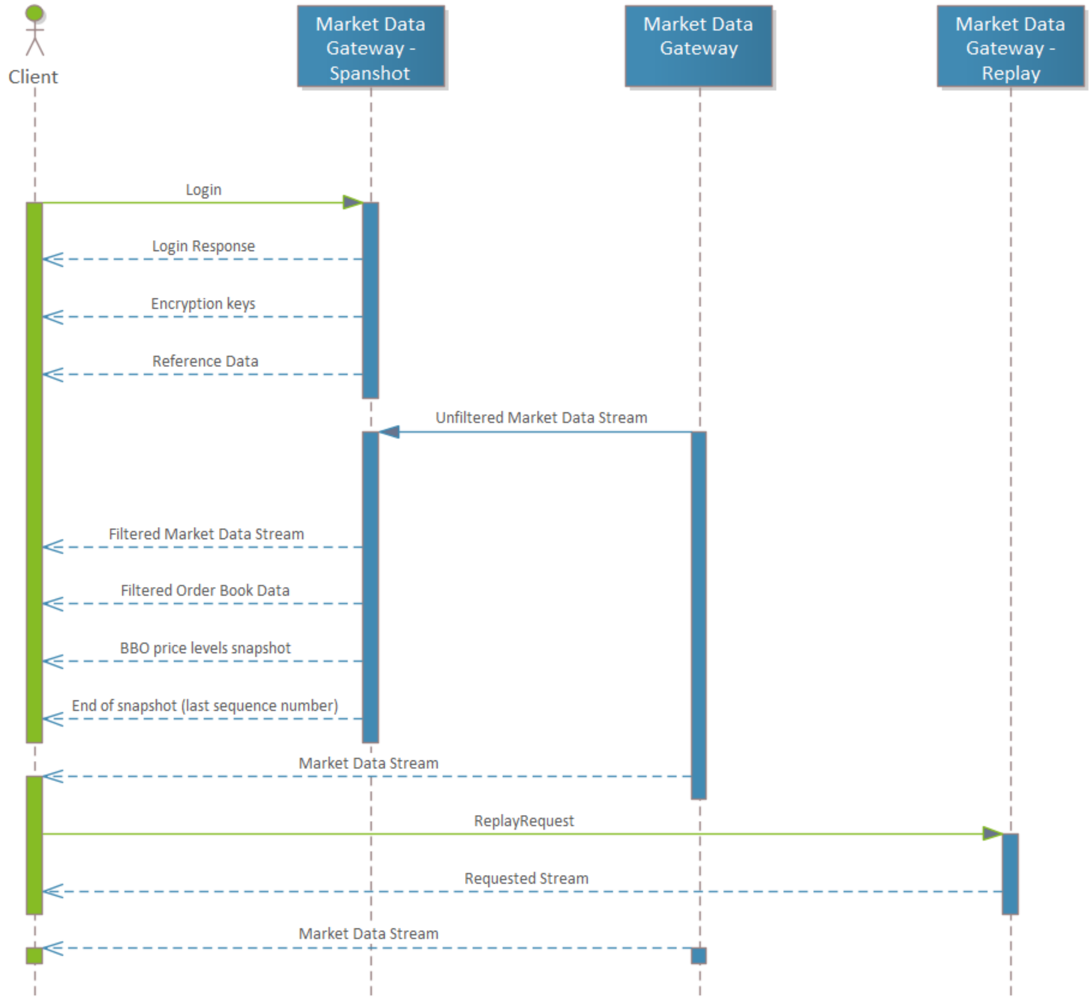
Upon connecting to the Snapshot, the user receives reference data and the full market state up to that point in time. The user then switches to the Stream service for real time data.

Replay is to be used when gaps in Stream are detected. If there are communication gaps between Snapshot and the real time Stream, a request for the missing data can also be sent to the Replay service.

The dedicated message is ReplayRequest, in which the requested data needs to be specified by a start and end sequence number. In return, the Replay service returns the missing data.

Channel	Delivery method
Stream	Realtime feed, UDP (multicast transmission)
Snapshot	On demand, TCP (unicast transmission)
Replay	On demand, TCP (unicast transmission)

Figure 2 - General Information Flow



4.3.2. SNAPSHOT

The Snapshot service is available via TCP (unicast transmission) for a given instance of Market Data and is implemented as a standalone service.

Snapshot is the first channel users connect to. It conveys reference data such as information about the structure of the market and the instruments which can be traded as well as schedules. In addition, it disseminates encryption keys for decrypting the information sent through Stream and Replay. Snapshot can be accessed at any time during the trading day to obtain aggregate information about the state of the order book and the full record of trades and transactions up to that time.

Snapshot:

- provides reference data and encryption keys for decrypting the information sent though Stream and Replay. It can be accessed at any time during the trading day to obtain the full depth order book and the complete record of trades and transactions,
- is cumulative. It does not communicate all the OrderAdd, OrderModify and OrderDelete messages sent to the system but only their cumulative effect up to the time the snapshot is requested.

The Snapshot is personalized for each user and only sends the data for which the user holds an active subscription. This includes a selection of Instruments, the choice of BBO or Full Depth data and other parameters.

4.3.3. STREAM

Stream is the real time channel for market data and disseminates order and transaction data over multicast UDP feeds. It publishes among others the full-depth order book, BBO levels as well as auction related statistics.

Stream is designed to operate with minimum technical delay. Publication is executed in a push model, and messages are published on the corresponding multicast group. The user application is responsible for receiving messages from the corresponding multicast group.

4.3.4. REPLAY

Gap detection is an extremely important mechanism for UDP transmissions to deliver complete information reliably and confidently to the user. The purpose of the Replay service is to retransmit message gaps to user applications which missed a part of the Stream.

Replay's mechanism recovers information lost by Stream during a multicast transmission and supports retransmitting of all Stream message types. It is implemented by the ReplayRequest message.

5. Real Time Flow In Market Data

5.1. Message groups

GPW WATS Market Data consists of sets of messages, which taken together enable a market participant to obtain a marketplace structure picture, as well as the flow of transactions in instruments. The table below shows the list of messages involved in Market Data broken down by message category. Details of the messages will be elaborated in the following sections and chapters.

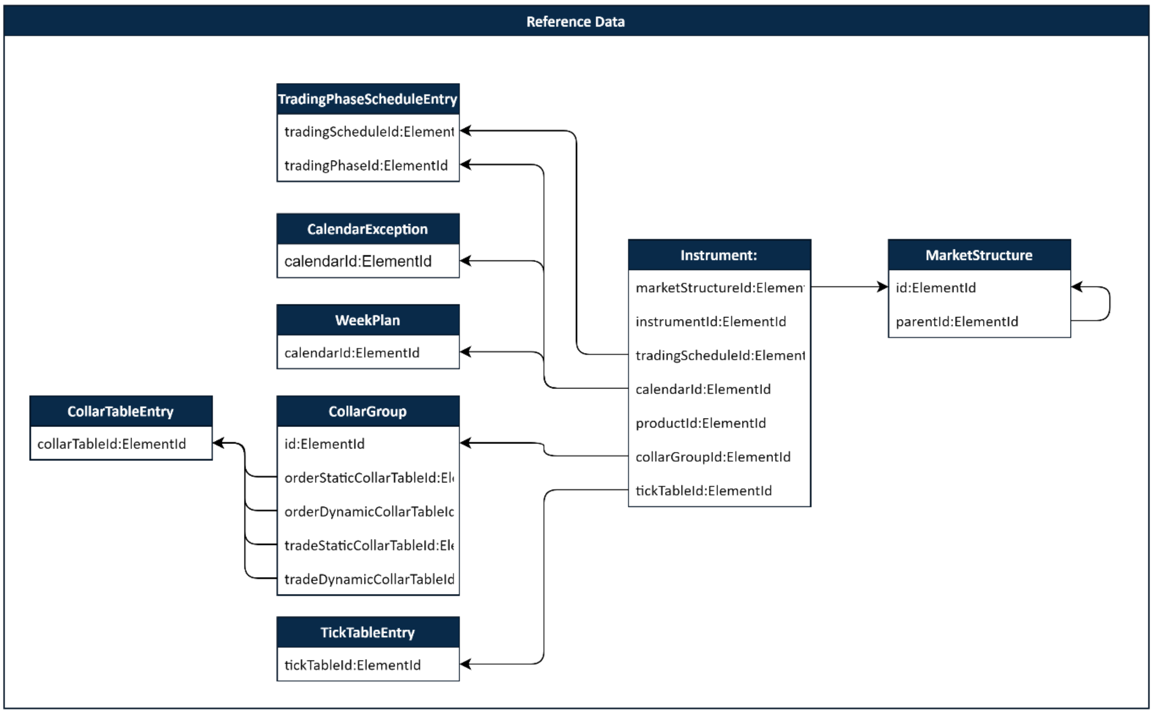
Category	Message type	Message	Description
Trading and execution	Auction	AuctionUpdate AuctionSummary	Messages which communicate trading activity. This includes messages in support of Auctions. This includes notification of executions and trades. BBO Levels are conveyed by PriceLevelSnapshot, the Full Depth order book by Order* and Auction time data by Auction* messages.
	Continuous Trading (Full Depth)	OrderAdd OrderDelete OrderModify OrderBookEvent OrderExecute	
	BBO	PriceLevelSnapshot	
Status messages	Status	InstrumentStatusChange PriceUpdate OrderCollars/TradeCollars Trade News OpenPositions	Messages that communicate the status of the market.
Summary messages	Statistical	InstrumentSummary SessionSummary	Summary statistics about instruments and trading sessions
Reference Data	Market Structure	MarketStructure Instrument TickTableEntry	Group of messages which define Market, instruments, and other market parameters.
	Trading Calendar	TradingPhaseScheduleEntry CalendarException WeekPlan	
	Order and Trade Collars	CollarTableEntry CollarGroup	
	Pricing Messages	AccruedInterestTableEntry IndexationTableEntry	Information about indexation and accrued interest for fixed income instruments
Index	Index messages	IndexParams IndexPortfolioEntry RealTimeIndex IndexSummary	Group of messages which power the Index feature.
Session Management	Encryption	EncryptionKey	Distribution of encryption keys.
	Technical	StartOfTechnicalSession EndOfTechnicalSession ReferenceDataStart ReferenceDataEnd EndOfSnapshot EncryptionKey Test and TestEvent ReplayRequest	Support messages used to frame the contents of Market Data.

5.2. The Big Picture

The diagrams in this section show the extent of inter-relations between different Market Data messages and the way they combine into a coherent whole. The graphs only feature the ID fields (keys) of the individual Market Data messages. The messages carry many more attributes than shown in this illustration. Session management messages together with the EncryptionKey message are omitted.

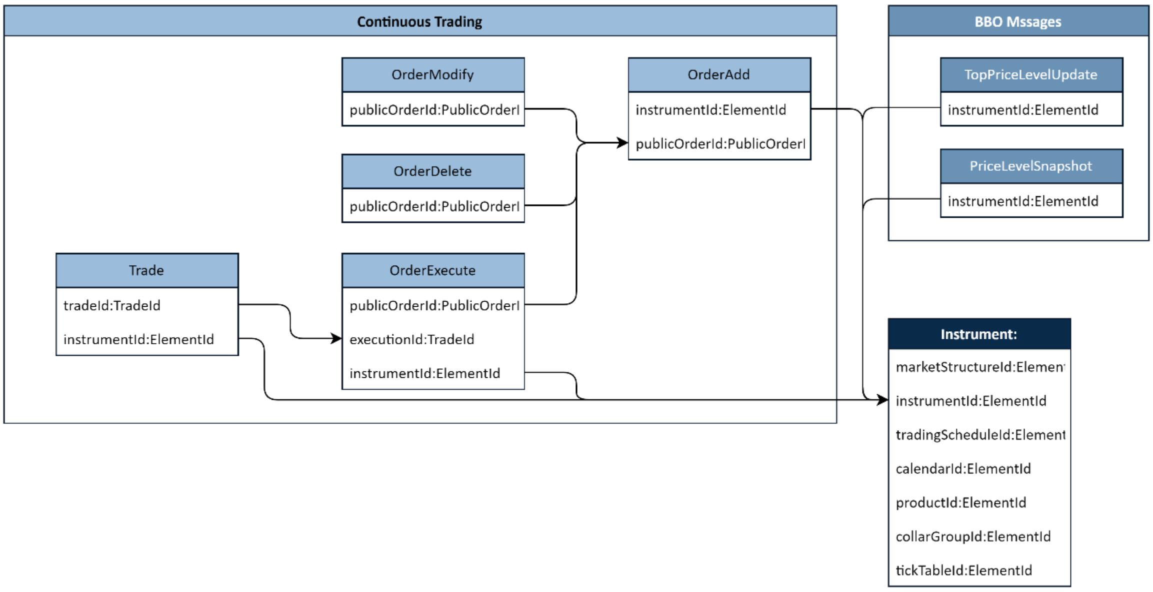
5.2.1. REFERENCE DATA

Figure 3 - Reference data



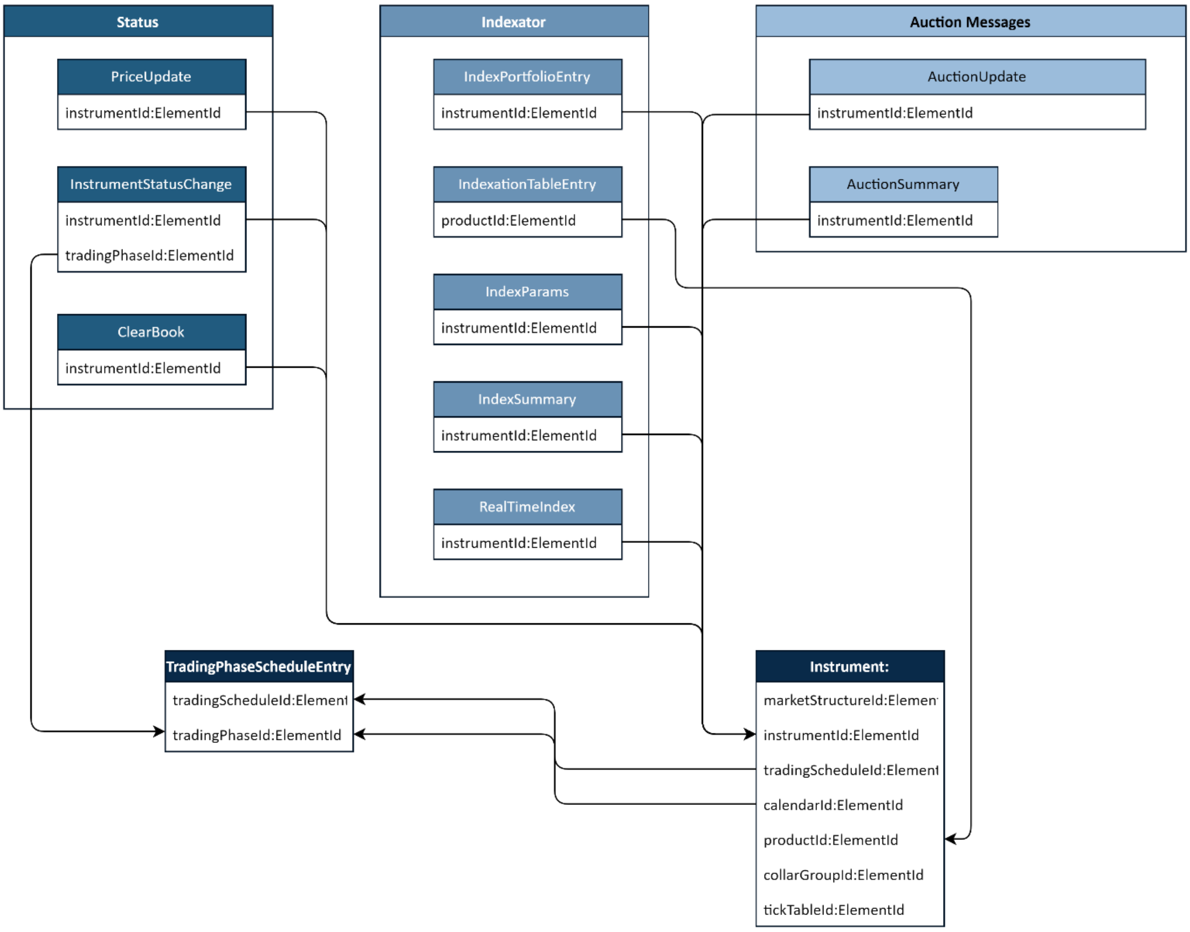
5.2.2. ORDER MESSAGES AND REFERENCE DATA

Figure 4 - Order messages and reference data



5.2.3. STATUS, AUCTION, INDEX MESSAGES AND REFERENCE DATA

Figure 5 - Status, Auction, Index Messages and Reference Data



5.3. Stream+Replay Contents: Full Depth and BBO

The real time Full Depth and BBO data share many of the same messages and differ in specific content. Full Depth conveys complete market information, including Auctions. BBO features only bid offer levels and omits OrderAdd, OrderModify, OrderDelete and OrderExecute messages.

The messages and categories of messages involved in the Full Depth Stream and the Full Depth Replay services.

Category	Full Depth Stream + Replay Messages	BBO Stream + Replay Messages
Auction	OrderBookEvent AuctionSummary AuctionUpdate	AuctionSummary AuctionUpdate
Continuous trading	OrderAdd OrderDelete OrderModify	
BBO		PriceLevelSnapshot
Execution and Trade	OrderExecute Trade OpenPositions	Trade OpenPositions
Trading Phase	InstrumentStatusChange	InstrumentStatusChange
Trading Collar	OrderCollars/TradeCollars	OrderCollars/TradeCollars *
Session Management	StartOfTechnicalSession EndOfTechnicalSession	

* optional, based on BBO service level configuration
** optional, can be configured at the system level

5.3.1. GENERAL SEQUENCE OF MARKET DATA MESSAGES DURING A TRADING DAY

Stream follows a predetermined structure during a Market Data trading day. The Stream operates within a technical session during which all market data is disseminated. At the beginning of the technical session, reference data is disseminated to all connected users. After this, once the trading day begins and orders start coming in, market messages are disseminated according to the rules for different trading phases. Once the technical session ends, Stream ceases publication of market data.

It is important to distinguish the technical session which is the framework for Market Data and a trading session. A trading session occurs within a technical session.

Message Type	Sequence of messages in Stream	Note
Session Management	StartOfTechnicalSession	The technical session starts. The GPW WATS system Stream is open and Market Data messages start flowing.
	ReferenceDataStart	Note that the Reference Data messages may arrive in random order and not in related contiguous chunks.
Reference Data		Reference data. For information on the structure of the reference data block, please see Section 5.4.
	ReferenceDataEnd	
Order Reference Data		OrderAdd messages for persistent orders
	Start of Trading Session	
Trading	AuctionUpdate AuctionSummary OrderAdd OrderDelete	Once the trading session starts, trading messages are published as they arrive. The publication of trading messages adheres to the rules of the trading phases, such as auctions and continuous trading. Market operations related messages such as InstrumentStatusChange and OrderCollars/TradeCollars set the

Message Type	Sequence of messages in Stream	Note
	OrderModify PriceLevelSnapshot OrderExecute Trade InstrumentStatusChange OrderCollars/TradeCollars	flow of information and convey data related to order control. The list represents a series of possible messages, and not their order of arrival.
	End of Trading Session	
Summaries	InstrumentSummary SessionSummary IndexSummary OpenPositions	Summaries and statistics about the trading session.
Session Management	EndOfTechnicalSession	The technical session ends. The Flow of Market Data messages is stopped.

5.3.2. REPLAY

The Replay service can re-transmit any of the messages from the Stream service which were lost during transmission. For each Stream the corresponding replay service, Replay Full Depth or Replay BBO, will retransmit the lost messages upon request. The list of messages which can be replayed is the same as in the table above.

5.4. Snapshot: Full Depth, BBO and Reference Data

5.4.1. SNAPSHOT LEVELS

The Snapshot service offers 3 levels of Snapshots, each customized to the needs of potential users. The Snapshots contain a core of reference data information but differ in the extent of market data provided. Here are the principal differences:

- Full Depth snapshot, aimed at users who rely on the full picture of the market for their trading and investment activities. It is the most complete option available and contains Reference data, the full depth order book and auction related information.
- BBO Snapshot, aimed at users who only require best bid-offer levels. It is otherwise identical to the Full Depth snapshot.
- Reference Data snapshot, for users who do not require trading data, but only information about the structure of the market.

5.4.2. SNAPSHOT SEQUENCE

The Snapshot disseminates critical market structure and whole-day trading information at the start of the trading day as well as during the day upon request. The sequence of the messages in the Snapshot is organized in groups of messages. Together these messages can help the user construct an image of the market structure and build an image of the order book. The structure of the Snapshot for Full Depth and BBO differs due to the scope of the data and is presented in the table below.

Message Type	Message sequence in Full Depth Snapshot	Message sequence in BBO Snapshot	Message sequence in Reference Data Snapshot
Encryption	EncryptionKey	EncryptionKey	EncryptionKey
Session Managemen	StartOfTechnicalSession	StartOfTechnicalSession	StartOfTechnicalSession
Session Management	ReferenceDataStart	ReferenceDataStart	ReferenceDataStart
Trading Calendar	WeekPlan	WeekPlan	WeekPlan
Trading Calendar	CalendarException	CalendarException	CalendarException
Market Structure	TickTableEntry	TickTableEntry	TickTableEntry
Trading Collar	CollarTableEntry	CollarTableEntry	CollarTableEntry
Trading Calendar	TradingPhaseScheduleEntry	TradingPhaseScheduleEntry	TradingPhaseScheduleEntry
Market Structure	MarketStructure	MarketStructure	MarketStructure
Pricing	AccruedInterestTableEntry	AccruedInterestTableEntry	AccruedInterestTableEntry
Trading Collar	CollarGroup	CollarGroup	CollarGroup
Market Structure	Instrument	Instrument	Instrument
Session Management	ReferenceDataEnd	ReferenceDataEnd	ReferenceDataEnd
Status	InstrumentStatusChange OrderCollars	InstrumentStatusChange OrderCollars	InstrumentStatusChange OrderCollars
Continuous Trading / BBO	OrderAdd Trade	PriceLevelSnapshot	
Session Management	EndOfSnapshot	EndOfSnapshot	EndOfSnapshot

6. Trading Messages

6.1. Auctions

6.1.1. AUCTIONUPDATE

AuctionUpdate is sent during an auction whenever IMP/IMV values change and may include the best buy and sell levels. As for an uncrossed book, only the best buy and sell levels are sent.

6.1.2. AUCTIONSUMMARY

Disseminated at the end of an auction after uncrossing. Contains the final auction price and quantity. The type of auction is specified by auctionType.

As for an uncrossed book, AuctionSummary sends zeros in the price and quantity fields.

6.1.3. ORDERBOOKEVENT

OrderBookEvent is a multipurpose message. Among other things it instructs Participants to clear the order book in anticipation of an upcoming auction. Its particular use follows auction uncrossing. It services the following events:

eventType	
Clear	Clear order book
ResendStart	Order book resend start (resend will begin shortly)
ResendEnd	Order book resend end (resend is done)

6.1.4. SEQUENCE OF MESSAGES DURING AN AUCTION

Auctions are announced and preceded by a special sequence of messages and InstrumentStatusChange is only one of the messages involved in the process. The following table outlines the sequence of messages leading up to an auction:

Sequence	Message	Category	Note
1	InstrumentStatusChange	Message sequence to set up the Auction.	Signals the change of trading phase to Auction.
2	OrderBookEvent		OrderBookEvent with eventType=Clear signals the need to clear the book in preparation for an auction. Auction starts and participants enter orders on the trading port.
3	OrderCollars/TradeCollars		OrderCollars/TradeCollars messages arrive in series to define dynamic or static collars.
4	AuctionUpdate	Auction Call	Informs participants on the state of the Auction Call.
5	AuctionSummary	Auction Uncrossing	Auction price publication after Uncrossing.
6	InstrumentStatusChange	Change of trading phase	Switching into another trading phase after Uncrossing (Continuous Trading or Auction). Enriched Trade reports from the auction follow this message. Please note that no Execution reports (OrderExecute) are generated for auctions.
7	OrderBookEvent		OrderBookEvent with eventType= ResendStart

Sequence	Message	Category	Note
		Post-auction book rebuilding process	This message functions as a signal to prepare rebuilding the order book and is followed by OrderAdd messages.
8	OrderBookEvent		OrderBookEvent with eventType= ResendEnd Orders related to rebuilding the order book have been sent. Trading can commence.

The messages surrounding an Auction for instrumentId=7 are reported in the table below. It is assumed that TradingPhaseScheduleEntry with tradingPhaseId=4 exists which identifies a tradingPhaseType="AuctionOpening" with a type auctionType="OpeningAuction" and that it is attached to an Instrument with InstrumentId=7.

Message	Message Contents
`InstrumentStatusChange`	`{"header":{"length":49,"msgType":"InstrumentStatusChange","version":4096,"seqNum":26050,"timestamp":1705476600008972078,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"tradingPhaseId":4,"status":"Active","stressedMarket":false}`
`OrderBookEvent`	`{"header":{"length":44,"msgType":"OrderBookEvent","version":4096,"seqNum":26056,"timestamp":1705476600009119282,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"eventType":"Clear"}`
`OrderCollars`	`{"header":{"length":84,"msgType":"OrderCollars","version":4096,"seqNum":26054,"timestamp":1705476600009042780,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"collarType":"Static","price":261000000,"lowerBid":130600000,"upperBid":391500000,"lowerAsk":130600000,"upperAsk":391500000}`
`TradeCollars`	`{"header":{"length":68,"msgType":"TradeCollars","version":4096,"seqNum":26052,"timestamp":1705476600008994278,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"collarType":"Dynamic","price":261000000,"lower":245500000,"upper":276500000}` `{"header":{"length":68,"msgType":"TradeCollars","version":4096,"seqNum":26053,"timestamp":1705476600009042780,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"collarType":"Static","price":261000000,"lower":209000000,"upper":313000000}`
`AuctionUpdate`	`{"header":{"length":107,"msgType":"AuctionUpdate","version":4096,"seqNum":26057,"timestamp":1705476600009132382,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"indicativeMatchingPrice":0,"indicativeMatchingVolume":0,"totalSellQuantityAtImp":0,"totalBuyQuantityAtImp":0,"bestSellLevel":0,"bestSellLevelQuantity":0,"bestBuyLevel":0,"bestBuyLevelQuantity":0}`

Message	Message Contents
`AuctionSummary`	{"header":{"length":60,"msgType":"AuctionSummary","version":4096,"seqNum":41080,"timestamp":1705478400003537367,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"price":0,"quantity":0,"auctionType":"OpeningAuction"}
`InstrumentStatusChange`	{"header":{"length":49,"msgType":"InstrumentStatusChange","version":4096,"seqNum":41081,"timestamp":1705478400003545268,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"tradingPhaseId":5,"status":"Active","stressedMarket":false}
`OrderBookEvent`	{"header":{"length":44,"msgType":"OrderBookEvent","version":4096,"seqNum":41083,"timestamp":1705478400003655271,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"eventType":"ResendStart"}
	`OrderAdd messages involved in book rebuilding.`
`OrderBookEvent`	{"header":{"length":44,"msgType":"OrderBookEvent","version":4096,"seqNum":41084,"timestamp":1705478400003728273,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"eventType":"ResendEnd"}

6.1.5. FOLLOWING AN AUCTION

At the beginning of a Continuous Trading phase following Auction, to enable users to reconstruct the order book, Market Data publishes a stream of order messages (OrderAdd) which represent the net order book from the beginning of the trading day. Market Data publishes OrderAdds for all orders left over from the auction and the market state at that time of the day. In addition, any orders triggered by the Auction price will be broadcast i.e., Stop Limit orders, except for Stop Loss orders.

6.1.6. AUCTION TIME SUPPRESSION

During Auctions, orders inbound on the trading port are suppressed on Market Data. Specifically, orders will be suppressed starting from the InstrumentStatusChange message which signals the start of the auction. Orders will restart following the end of an Auction phase which is communicated through a suitable InstrumentStatusChange message. The list of messages not published on Market Data during Auctions is:

Messages Suppressed	Note
OrderAdd **OrderDelete** **OrderModify**	These messages are not published during Auction Calls in line with the auction regulations.
OrderExecute **Trade**	These messages are not generated during auctions because during auction calls, transactions are not made. They are therefore not explicitly suppressed. OrderExecute is not broadcast during auction calls or uncrossing. Trade messages are published as normal after auction uncrossing to communicate the outcomes of the auction. The message sequence is: AuctionSummary, InstrumentStatusChange out of the Auction, OrderExecute and Trade messages.

6.1.7. EXAMPLES

6.1.7.1. AuctionUpdate

AuctionUpdate sends updates in the IMP/IMV or BBO during an Auction.


```
{"header":{"length":107,"msgType":"AuctionUpdate","version":4096,"seqNum":26057,"timestamp":1705476600009132382,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"indicativeMatchingPrice":0,"indicativeMatchingVolume":0,"totalSellQuantityAtImp":0,"totalBuyQuantityAtImp":0,"bestSellLevel":0,"bestSellLevelQuantity":0,"bestBuyLevel":0,"bestBuyLevelQuantity":0}
```


A sample update message during the auction call for instrumentId=7.

The behavior of the AuctionUpdate message is illustrated below:

Field name	Description	Behavior
instrumentId	A reference to the instrument.	
indicativeMatchingPrice	Indicative Matching Price and Indicative Matching Volume. Disseminated whenever a change in IMP/IMV occurs.	These fields are only filled in the case of a Crossed Book.
indicativeMatchingVolume		
totalSellQuantityAtImp	The buy and sell quantity.	
totalBuyQuantityAtImp		
bestSellLevel	BBO Level 1 components.	In the case of Uncrossed Book, only the first BBO level is broadcast.
bestSellLevelQuantity		
bestBuyLevel		
bestBuyLevelQuantity		

6.1.7.2. AuctionSummary

AuctionSummary is sent at the conclusion of an Auction and provides the results of the Auction Uncrossing.


```
{"header":{"length":60,"msgType":"AuctionSummary","version":4096,"seqNum":41070,"timestamp":1705478400003252460,"sourceTimestamp":1705478400003121256,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":4,"price":0,"quantity":0,"auctionType":"OpeningAuction"}
```


AuctionSummary is sent at the end of an auctionType=OpeningAuction, communicates the price and quantity arrived at during the auction for instrumentId=4.

6.2. Full Depth Order Messages

6.2.1. ORDERID AND PUBLICORDERID

GPW WATS uses the publicOrderId attribute to identify orders on Market Data. A publicOrderId is assigned to every order entered into the system.

GPW WATS assigns each order with two distinct attributes: orderId and publicOrderId. The publicOrderId identifies the order on Market Data while the orderId remains private for each trading party.

A trading party obtains these identifiers upon the successful submission of an order on the trading port by means of response messages. On the Native trading port an OrderAddResponse message will contain the orderId and publicOrderId corresponding to an order successfully submitted by the OrderAdd message. The FIX trading port follows a similar logic.

The Market Data publicOrderIds are reset daily and start each day with the value 1, each new order being assigned a sequentially increasing value. publicOrderIds are unchangeable during the trading day and are only valid within each trading session.

The Market Data publicOrderId cannot be used to modify or cancel an order. To modify or cancel an order, the orderId from the OrderAddResponse message needs to be employed.

6.2.2. ORDERADD

The message specifies the side, instrument (InstrumentId), (limit) price and quantity.

The publicOrderId referenced within the message corresponds to the publicOrderId received in the OrderAddResponse message in response to an OrderAdd on the Native or in the Execution Report message in response to a New Order Single on the FIX trading ports.

6.2.3. ORDERDELETE

The publicOrderId identifies the order on the Market Data stream.

6.2.4. ORDERMODIFY

The Price and Quantity fields represent the new values for the order. The PriorityFlag indicates whether the order retained or lost priority and is set following time-priority rules.

For further information about the impact of modifications on order priority in the CLOB, please refer to the GPW WATS 1.01 Trading System document.

6.2.5. ORDEREXECUTE

Execution Report specifies the execution price and quantity (executionQuantity). OrderExecute generates 2 reports, one for each side of the transaction and identified by different publicOrderIds. No transaction side information is provided. The publicOrderId refers to a Market Data OrderAdd message.

The Quantity field represents the quantity of the order left over in the Order Book. If Quantity=0, the order has been fully filled and can be removed from the Order Book.

The executionId in both OrderExecute messages is the same and refers to a unique TradeID in the Market Data Trade message.

OrderExecute messages are not published on the BBO Stream.

6.2.6. EXAMPLES

6.2.6.1. OrderAdd

OrderAdd informs that a limit order was added to the order book. It is sent immediately upon the entry of the order into the order book. The publicOrderId uniquely identifies the order for that given trading day.

```
{"header":{"length":68,"msgType":"OrderAdd","version":4096,"seqNum":56090,"timestamp":1705479629764875700,"sourceTimestamp":1705479629764762397,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"publicOrderId":1,"instrumentId":514,"side":"Sell","price":38100000000,"quantity":1}
```


A Sell order for Instrument 514 was entered in the system. The Order is identified by publicOrderId=1.

6.2.6.2. OrderModify and OrderDelete

OrderModify and OrderDelete are related to OrderAdd. They refer to previous orders by means of publicOrderId and InstrumentIDs. OrderModify informs that an order with publicOrderId was modified and provides the new price and quantity. In addition, it confirms if the order has lost or retained priority. OrderDelete reports that an order was deleted from the order book. The message refers to the deleted order through publicOrderId.


```
{"header":{"length":68,"msgType":"OrderModify","version":4096,"seqNum":29884,"timestamp":1699289311630633951,"sourceTimestamp":1699289311630280464,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":4672,"publicOrderId":19,"price":10000000000,"quantity":2000,"priorityFlag":"Lost"}
```


Order number 19's price was modified and lost priority. The new price and quantity are included.


```
{"header":{"length":51,"msgType":"OrderDelete","version":4096,"seqNum":29889,"timestamp":1699289313739959063,"sourceTimestamp":1699289313739478138,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":4669,"publicOrderId":22}
```


Order 22 was deleted.

6.2.6.3. OrderExecute and Trade

OrderExecute is sent in pairs, one for each side, when an order executes, and a trade is produced on the order book. The message refers to an order by the publicOrderId field. The execution may be full, in which case quantity=0, or partial. The ID of the related Trade is provided by executionId.

OrderExecute is published immediately upon a match. Trade differs from OrderExecute in that it is the official record of the transaction and appears with a slight delay due to its size and contents.


```
{"header":{"length":79,"msgType":"OrderExecute","version":4096,"seqNum":56180,"timestamp":1705484415023600494,"sourceTimestamp":1705484415023454890,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"quantity":20,"instrumentId":1669,"publicOrderId":21,"executionId":2,"executionPrice":760000000,"executionQuantity":2}
```


The first of two OrderExecute messages inform of an execution: order in instrument 1669 executed fully ("remaining" quantity=0) at a price of 760000000. The trade will be reported by the Trade message with tradeId=executionId=2.


```
{"header":{"length":153,"msgType":"Trade","version":4096,"seqNum":56183,"timestamp":1705484415023848901,"sourceTimestamp":1705484415023789699,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":166
```

```
9,"settlementValue":1520,"settlementDate":20240119,"tradingTimestamp":1705484415023454890,"publicProductIdentification":{"productIdentification":"PLPGER000010","productIdentificationType":"ISIN"},"price":760000000,"priceCurrency":"PLN","priceNotation":"Price","quantity":2,"nominalValue":1520,"nominalCurrency":"PLN","mic":"XWAR","publicationTimestamp":1705484415023454890,"tradeId":2,"tradeToBeCleared":true,"mmtMarketMechanism":"CentralLimitOrderBook","mmtTradingMode":"ContinuousTrading","mmtTransationCategory":"None","mmtNegotitationIndicator":"NoNegotiatedTrade","mmtAgencyCrossTradeIndicator":"NoAgencyCrossTrade","mmtModificationIndicator":"NewTrade","mmtBenchmarkReferencePriceIndicator":"NoBenchmarkorReferencePriceTrade","mmtSpecialDividendIndicator":"NoSpecialDividendTrade","mmtOffBookAutomatedIndicator":"UnspecifiedOrNotApply","mmtOrdinaryTradeIndicator":"PlainVanillaTrade","mmtAlgorithmicIndicator":"AlgorithmicTrade","mmtPostTradeDeferralReason":"ImmediatePublication","mmtPostTradeDeferralType":"NotApplicable"}
```


A Trade tradeId=2 was made in instrument 1669 for a quantity of 2 at a price of 760000000.

6.2.6.4. OrderBookEvent

OrderBookEvent can signal the need to clear an Instrument's order book for an upcoming Auction.


```
{"header":{"length":44,"msgType":"OrderBookEvent","version":4096,"seqNum":41073,"timestamp":1705478400003278461,"sourceTimestamp":1705478400003121256,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":4,"eventType":"Clear"}
```


This OrderBookEvent message instructs participants to clear the order book for an upcoming Auction in the Instrument instrumentId=4.

6.3. BBO Messages

6.3.1. PRICELEVELSNAPSHOT AND TOPPRICELEVELUPDATE

PriceLevelSnapshot and TopPriceLevelUpdate are specific to the BBO dataset.

They contain the Best Bid and Offer data such as prices, quantities, and order counts for a security (InstrumentID) at any given time.

PriceLevelSnapshot contains up to 10 BBO levels while TopPriceLevelUpdate contains only the best bid and offer.

The depth of the BBO is defined by the Market operator. By default, GPW WATS provides 10 BBO levels (parameter maxDepth). The data encrypted for each BBO level is price, quantity and OrderCount.

Each BBO level is encrypted with a different encryption key (encryptionKeyId). The keys required to read the BBO levels are disseminated by the Snapshot service EncryptionKey message. To decrypt a particular BBO level, the user must employ the key with the relevant encryptionKeyId.

6.3.2. SEQUENCE OF MESSAGES ON BBO DURING AN AUCTION

The table below presents the general flow of BBO messages around auctions. At auction start, a PriceLevelSnapshot filled with zeros follows an InstrumentStatusChange but precedes AuctionUpdate messages. Inside auctions, PriceLevelSnapshots are muted and AuctionUpdate messages update BBO recipients on auction progress. AuctionSummary recaps auction results. PriceLevelSnapshot resume upon completion of auctions after rebuilding of the order book.

Sequence	Message	Category	Note
1	InstrumentStatusChange	Sequence of messages setting up the Auction.	Signals the change of trading phase to Auction.
2	OrderCollars/TradeCollars		OrderCollars/TradeCollars define dynamic or static collars.
3	PriceLevelSnapshot		PriceLevelSnapshot with zeros (0) in all fields signals the start of the auction.
4	AuctionUpdate	Auction Call	Updates on the state of the Auction.
5	AuctionSummary	Auction Uncross	Auction price publication after Uncrossing.
6	InstrumentStatusChange	Change of trading phase	Switching into another trading phase after Uncrossing (Continuous Trading or Auction). Enriched Trade reports from the auction follow this message.
7	PriceLevelSnapshot	Post-auction	PriceLevelSnapshots resume after the auction.

6.3.3. BBO MESSAGE SCHEDULE

Unlike Order Messages (OrderAdd, OrderDelete, and OrderModify), BBO messages are not disseminated in real time. The rules for broadcasting BBO messages are as follows:

- PriceLevelSnapshot is generated at regular time intervals measured in milliseconds. PriceLevelSnapshot contains all the changes to price levels which occurred since its last publication. The publication interval is currently set at 100ms and can be configured at the system operator level. The recommended practice is to choose a delay of 50-100ms.
- PriceLevelSnapshot can be configured to send either the best bid and offer, or 5 or 10 BBO levels.

6.3.4. EXAMPLES

6.3.4.1. PriceLevelSnapshot

The PriceLevelSnapshot communicates 10 BBO levels in the form of price, quantity and orderCount. The BBO depth is set by the maxDepth parameter, and in the following message it is 2. PriceLevelSnapshot is sent out at regular intervals of 100ms.


```
{"header":{"length":664,"version":0,"msgType":"PriceLevelSnapshot","seqNum
":22538,"timestamp":"2023-03-28
11:34:13.439440733","sourceTimestamp":"2023-03-28
11:34:13.410688751","isEncrypted":false,"encryptionOffset":0,"encryptionKe
yId":0},"instrumentId":145,"maxDepth":2,
"buy":[{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"pric
e":10000000000,"quantity":7,"orderCount":1},{"isEncrypted":false,"encrypti
onOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"i
sEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quan
tity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encrypt
ionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"e
ncryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount"
:0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":
0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"
encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":f
```

```
alse,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orde
rCount":0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"
price":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffs
et":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0}],"sell":[
{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"q
uantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encr
yptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":false
,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCou
nt":0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"pric
e":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":
0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted
":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"o
rderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":
0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionO
ffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEn
crypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantit
y":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encryption
KeyId":0,"price":0,"quantity":0,"orderCount":0}]}
```


Instrument 145, on 2023-03-28 at 11:34:13.439440733, featured only one set of prices on the buy side price=10000000000, quantity=7, orderCount=1.

6.4. Hybrid Market

Market Data plays a role in the functioning of the Hybrid Market.

6.5. Status Messages

Status change messages signal important events in the state of the market. They comprise of:

Message	Description
InstrumentStatusChange	Signals the change of Trading Phase for a given Instrument. May indicate stressed market conditions.
PriceUpdate	Communicates the new reference price for an Instrument.
OrderCollars	Communicates calculated Order price collars for an Instrument.
TradeCollars	Communicates calculated Trade price collars for an Instrument.

6.5.1. INSTRUMENTSTATUSCHANGE

InstrumentStatusChange signals that an Instrument (InstrumentID) has entered a new trading phase (tradingPhaseId).

A secondary role of the message is to communicate Stressed Market Conditions for particular instruments. This is accomplished by means of the stressedMarket=true/false flag.

InstrumentStatusChange refers to the trading phase defined by the TradingPhaseScheduleEntry message via the tradingPhaseId field. TradingPhaseScheduleEntry is part of the Reference Data disseminated by the Snapshot service.

Instruments can be in the states described in the table below. Only one state is in force at any given time. The Market Operations team has the responsibility of setting and releasing instruments from states.

Name	Description
Active	Trading is allowed. Order placement and trades may take place as long as the trading phase allows for such activity.
Inactive	Trading is not allowed. Any operations aimed at order handling including order placement, modification or cancellation are not allowed.
MarketOperationsSuspension	Manual suspension by Market Operations in case of an extraordinary event impacting trading.
OutsideCollarsStatic / OutsideCollarsDynamic	Activated automatically whenever an attempt is observed to conclude a trade outside the collars (order or trade collars, when applicable). Once the status is triggered, a Volatility Auction takes place.
RegulatorySuspension	Reserved for Financial Supervisory Authority decisions.
TechnicalHalt	Triggered in cases of system failure or technical errors related to an instrument. The event causes the order book to be closed and cleared.

6.5.2. PRICEUPDATE

PriceUpdate communicates the new reference price for an Instrument (InstrumentId).

The price can be a reference price or a midpoint price. The "price" field represents the value of the reference price at the time of the message.

6.5.3. ORDERCOLLARS AND TRADECOLLARS

OrderCollars and TradeCollars communicate the calculated values of the order and trade price collars for a particular security (InstrumentId) based on the ranges defined in the CollarGroup message.

The "price" field represents the value of the reference price used to compute the collars while "collarType" describes whether the collar in question is static or dynamic.

Collar values are contained in the remaining parts of the message.

Message	Fields	Note
OrderCollars	lowerBid, upperBid lowerAsk, upperAsk	OrderCollars disseminates the lower and upper bid and ask limits for order entry
TradeCollars	lower upper	TradeCollars disseminates the lower and upper bounds within which trades can be made

The functioning of price collars is explained in more detail in the **GPW WATS 1.01 Trading System** document.

6.5.4. EXAMPLES

6.5.4.1. InstrumentStatusChange - Switching between Phases

When an instrument switches between scheduled (or unscheduled) phases, an InstrumentStatusChange message is broadcast on Market Data. In the example below, a Closing Auction is scheduled to take place from 15:50 UTC to 16:00 UTC. At 16:00 UTC an InstrumentStatusChange will be sent on Market Data to signal the change of phases and at 16:00 UTC the Instrument which follows this schedule (tradingScheduleId=1) will switch to Trade at Last.

```
{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":4096,"seqNum":231,"timestamp":1705460401082055566,"sourceTimestamp":1705460401082021465,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tradingScheduleId":1,"tradingPhaseId":6,"tradingPhaseSettlementCycle":2,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":10,"dynamicCollarVolatilityAuctionId":27,"tradingPhaseStartTime":1705506600000000000,"tradingPhaseStopTime":1705507200000000000,"tradingPhaseType":"AuctionClosing","auctionType":"ClosingAuction","auctionUncrossing":true,"extStaticCollarVolatilityAuctionId":11,"extDynamicCollarVolatilityAuctionId":28}

{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":4096,"seqNum":232,"timestamp":1705460401082165168,"sourceTimestamp":1705460401082138168,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tradingScheduleId":1,"tradingPhaseId":7,"tradingPhaseSettlementCycle":2,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":10,"dynamicCollarVolatilityAuctionId":27,"tradingPhaseStartTime":1705507200000000000,"tradingPhaseStopTime":1705507500000000000,"tradingPhaseType":"ContinuousLastAuctionTime","auctionType":"NotApplicable","auctionUncrossing":false,"extStaticCollarVolatilityAuctionId":11,"extDynamicCollarVolatilityAuctionId":28}
```


Two TradingPhaseScheduleEntry messages define 2 Trading Phases tradingPhaseId={6,7} which are part of the same Trading Schedule, tradingScheduleId=1. Trading Phase tradingPhaseId=6 is a Closing Auction and lasts from 15:50H to 16:00H, tradingPhaseId=7 is Trade at Last and lasts from 16:00H to 16:05H.

The relevant InstrumentStatusChange change message which signals the transition between the two phases refers to the trading phase, which is about to begin, tradingPhaseId=7. It signals that Instrument instrumentId=240 enters the new trading phase.


```
{"header":{"length":49,"msgType":"InstrumentStatusChange","version":4096,"seqNum":105448,"timestamp":1705507200096882261,"sourceTimestamp":1705507200096796559,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":240,"tradingPhaseId":7,"status":"Active","stressedMarket":false}
```


Instrument 240 enters Trading Phase 7. The transition occurs at UTC Wed Jan 17 2024 16:00:00.

6.5.4.2. PriceUpdate

PriceUpdate communicates the reference price applicable to given Instrument.


```
{"header":{"length":101,"version":4096,"msgType":"PriceUpdate","seqNum":23299,"timestamp":"2023-02-10T12:09:59.191830Z","sourceTimestamp":"2023-02-10T12:09:59.188603Z","isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":155,"priceType":"ReferencePrice","price":10000000000}
```


This PriceUpdate reports that the value of the reference price for Instrument 155 (`instrumentId=155`) is `price=100`. It is a reference price, and not a midpoint price.

6.5.4.3. OrderCollars and TradeCollars

OrderCollars and TradeCollars communicate values of order and trade collars for a given Instrument based on the ranges defined in CollarTableEntry.

When an Instrument features both Dynamic and Static collars, 2 OrderCollars/TradeCollars messages appear on Market Data, one for each type. When only one collar type is configured, one OrderCollars/TradeCollars message (either static or dynamic) appears. Whether 1 or 2 OrderCollars/TradeCollars messages are sent depends on the configuration of the Instrument.


```
{"header":{"length":84,"msgType":"OrderCollars","version":4096,"seqNum":12
3894,"timestamp":1705507500206244876,"sourceTimestamp":1705507500206120973
,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrument
Id":1489,"collarType":"Static","price":595000000,"lowerBid":298000000,"upp
erBid":890000000,"lowerAsk":298000000,"upperAsk":890000000}
```


This OrderCollars reports that the value of the reference price for Instrument 1489 (instrumentId=1489) is price=595000000.
The type of collars in the message is Static (collarType=Static).
The range of Static Buy Order Collars in the currency of Instrument 1407 is [298000000, 890000000]
The range of Static Sell Order Collars in the currency of Instrument 1407 is [298000000, 890000000]
These collar values and are expressed in a unit of the instrument's trading currency, here assumed PLN.
This is only one of two possible OrderCollars messages for Instrument 1489. If configured with 2 collar types, collarType=Dynamic and collarType=Static, a second OrderCollars would also appear.


```
{"header":{"length":68,"msgType":"TradeCollars","version":4096,"seqNum":57
88,"timestamp":1705460401848694429,"sourceTimestamp":1705460401848434122,"
isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId
":1489,"collarType":"Static","price":595000000,"lower":476000000,"upper":7
10000000}
```


TradeCollars reports that the value of the reference price for Instrument 1489 (instrumentId=1489) is price=595000000. TradeCollars sends the values of the collars, and not their deviations as in the case in CollarTableEntry.
The type of collars in the message is Static (collarType=Static).
The value of Static Trade Collars in the trading currency of Instrument 1489 is [476000000, 710000000]
These ranges are collar values and are expressed in a unit of the instrument's currency, here assumed PLN.
This is only one of two possible TradeCollars messages for Instrument 1489. If configured with 2 collar types, collarType=Dynamic and collarType=Static, a second TradeCollars would also appear.

6.5.5. TRADE

Trade is an enriched message which sends trade-related information after the order is processed. Trade messages include MIFID flags which follow the MMT specification. The TradeID attribute refers to the original OrderExecute message. Trade messages are also published on the BBO Stream.


```
{"header":{"length":153,"msgType":"Trade","version":4096,"seqNum":56145,"t
imestamp":1705480949016060026,"sourceTimestamp":1705480949015964823,"isEnc
rypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":169
0,"settlementValue":3601,"settlementDate":20240119,"tradingTimestamp":1705
480949015611714,"publicProductIdentification":{"productIdentification":"PL
PKO0000016","productIdentificationType":"ISIN"},"price":3601000000,"priceC
urrency":"PLN","priceNotation":"Price","quantity":1,"nominalValue":3601,"n
```

```
ominalCurrency":"PLN","mic":"XWAR","publicationTimestamp":1705480949015611
714,"tradeId":1,"tradeToBeCleared":true,"mmtMarketMechanism":"CentralLimit
OrderBook","mmtTradingMode":"UnscheduledAuction","mmtTransationCategory":"
None","mmtNegotitationIndicator":"NoNegotiatedTrade","mmtAgencyCrossTradeI
ndicator":"NoAgencyCrossTrade","mmtModificationIndicator":"NewTrade","mmtB
enchmarkReferencePriceIndicator":"NoBenchmarkorReferencePriceTrade","mmtSp
ecialDividendIndicator":"NoSpecialDividendTrade","mmtOffBookAutomatedIndic
ator":"UnspecifiedOrNotApply","mmtOrdinaryTradeIndicator":"PlainVanillaTra
de","mmtAlgorithmicIndicator":"NoAlgorithmicTrade","mmtPostTradeDeferralRe
ason":"ImmediatePublication","mmtPostTradeDeferralType":"NotApplicable"}
```


A trade was made in instrument 1690 for 1 unit at a price of PLN 3601000000.

6.5.6. NEWS

The News message enables the market operator to send messages addressed to a specific market. The market is identified by a marketStructureID from the MarketStructure message. The body of the message consists of a title and a text string and can be up to 800 characters long.

6.5.7. OPENPOSITIONS

The OpenPositions message communicates the number of open positions in a financial instrument.

The OpenPositions message may be sent out at predetermined time intervals as well as following each transaction on a derivative instrument. The publication frequency varies by source of the data.

The first publication of OpenPositions takes place before session start and continues during trading.

6.6. Summary Messages

6.6.1. INSTRUMENTSUMMARY

InstrumentSummary presents various statistical information about an instrument's trading session (instrumentId). The message is sent at the end of each trading session after market close. If a corporate action is planned for an instrument, the message is distributed once all necessary information is collected.

Name	Description
lastTradedPrice	Last Traded Price (LTP)
closingPrice	Closing Price (CP)
closingPriceType	Closing Price Type: LTP - Last Traded Price, LastACP – Last Adjusted Closing Price, FairValue, DailySettlementPrice, FinalSettlementPrice
adjustedClosingPrice	Adjusted Closing Price (ACP).
adjustedClosingPriceReason	Reason for adjusting the Closing Price. Potential reasons: Regular, Dividend, Issue Right, Split, Reverse Split, Bonus, Spin Off, Tick Table Change, Multiple Corporate Actions
pctChange	Percentage change from last session closing price
VWAP	Volume-weighted average price
noTrades	Total number of transactions in the current trading day
totalVolume	Total transaction volume
totalValue	Total transaction value
openPrice	The price of the first trade in the current trading day

Name	Description
maxPrice	Highest price of the instrument in the current trading day
minPrice	Lowest price of the instrument in the current trading day


```
{"header":{"length":133,"msgType":"InstrumentSummary","version":4096,"seqNum":119700,"timestamp":1705507500002469476,"sourceTimestamp":1705507500001982963,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":1,"lastTradedPrice":478000000,"closingPrice":478000000,"closingPriceType":"LTP","adjustedClosingPrice":478000000,"adjustedClosingPriceReason":"Regular","pctChange":10622222222,"vwap":478971979,"noTrades":5402,"totalVolume":27301,"totalValue":13076414,"openingPrice":480000000,"maxPrice":480000000,"minPrice":478000000}
```


An example InstrumentSummary message for instrumentID=1

6.6.2. SESSIONSUMMARY

SessionSummary is available to recipients with a valid subscription and requisite encryption keys.

It presents end-of-day summary statistics for an instrument. While the contents of this message overlap to some extent with those of InstrumentSummary, the messages are fundamentally different. SessionSummary contains enriched summary information calculated from trading day statistics for a given instrument.

7. Reference Data Messages

This chapter outlines the set of messages relayed in Reference Data and used to set up the market trading environment.

7.1. Market Structure Messages

7.1.1. MARKETSTRUCTURE

MarketStructure messages combine to create a market structure for a Market. The market structure is defined as a tree and starts with specifying a top-level element and continues by defining market segments. Multiple Markets may be defined simultaneously. Market segments are identified by a Market Identifier Code (MIC) as specified in ISO 10383, named and assigned a market model type.

Instruments (InstrumentId) are the leaves on the market structure tree. The market structure is populated with instruments from Instrument messages.

7.1.2. INSTRUMENT

Instrument defines an instrument that can be traded on the GPW WATS platform (InstrumentId).

The message contains the configuration for a security.

7.1.3. TICKTABLEENTRY

TickTableEntry messages combine to produce a tick table which may be assigned to an instrument. A Tick table starts at 0 and specifies the tick sizes for various price levels. lowerBound specifies the price level starting from which a tickSize applies.

7.1.4. EXAMPLES

Market Data broadcasts the market structure by sending a series of messages to that effect. A market structure is a set of trees, each of which defines a Market. A Market is a tree with branches which are market segments, sub-segments, or groups. The leaves on the branches of a Market tree are Instruments.

Instruments are collections of essential information which enable a product to trade. An Instrument incorporates many of the components defined by Reference Data messages (such as trading schedules, collars, tick tables etc.) into a single object. Instruments are allocated to Market Structures.

7.1.4.1. Market Structure

A Market Structure is a series of MarketStructure messages which define a tree of trading groups, segments, and sub-segments. An Instrument makes reference to a market structure branch of which is a leaf.


```
{"header":{"length":102,"msgType":"MarketStructure","version":4096,"seqNum
":278,"timestamp":1705460401088313131,"sourceTimestamp":170546040108828273
0,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"id":1,"pa
rentId":0,"mic":"XWAR","name":"WSE RM
Equities","marketModelType":"NotApplicable"}
```

An element of market structure has its own id, a parentId, features a MIC code, and a type qualifier structureType. parentId=0 indicates that this particular market element is the root of the market structure tree The name of the venue defined in this message is WSE RM Equities.

The following table describes details of the MarketStructure message in more detail.

Field name	Description	Message 278 (seqNum=278)
id	The ID of the Market Structure element.	1
parentId	A reference to the parent Market Structure element, i.e. the id field of the parent structure. The root of the structure has parentId=0.	0
mic	Market Identifier Code (MIC) of that particular element of the structure (ISO 10383). Can feature one or more MIC codes.	XWAR
name	The name of the element assigned by the Market Operator.	WSE RM Equities
marketModelType	The type of market structure element. The top element of any Market Structure is NotApplicable. Possible values for marketModelType include: 1. CLOB 2. BLOCK 3. HYBRID 4. CROSS 5. NotApplicable. The NotApplicable value is used in defining the top-level element of the market structure tree.	NotApplicable.

7.1.4.2. Instrument Definitions

Instruments are the nexus of Reference Data. They are defined using the Instrument message which binds all the Market Data reference messages together and defines its behavior by reference to them.

The key fields of interest in an Instrument message are as follows.

Field name	Refers to Messages [field]	Description
instrumentId		The ID of the Instrument.
productType		Type of financial product: 1. FinancialProductShare 2. FinancialProductBond 3. FinancialProductDerivativeFutures 4. FinancialProductDerivativeOptions 5. FinancialProductIndex 6. FinancialProductCurrency
calendarId	WeekPlan [calendarId] CalendarException [calendarId]	Trading calendar for the security.
marketStructureId	MarketStructure [id]	The branch of the Market Structure to which the Instrument belongs.
tickTableId	TickTableEntry [tickTableId]	The tick table for the instrument.
collarGroupId	CollarGroup [id]	The collar group.
tradingScheduleId	TradingPhaseScheduleEntry [tradingScheduleId]	The scheduled trading phases defined for the instrument.

The following messages refer to the Instrument through the instrumentId field:

Message Category	Message name
Order Messages	OrderAdd, OrderExecute, Trade Note: OrderModify and OrderDelete refer to OrderAdd only.

Auction Messages	Auctionupdate, AuctionSummary
BBO Messages	PriceLevelSnapshot
Status Messages	InstrumentStatusChange, OrderCollars and TradeCollars, OrderBookEvent


```
{"header":{"length":557,"msgType":"Instrument","version":4096,"seqNum":353
5,"timestamp":1705460401718556989,"sourceTimestamp":1705460401718330083,"i
sEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId"
:1,"productType":"FinancialProductShare","firstTradingDate":20070129,"last
TradingDate":0,"productId":360,"currency":"PLN","lotSize":1,"priceExpressi
onType":"Price","calendarId":1,"marketStructureId":10,"tickTableId":1,"col
larGroupId":3,"tradingScheduleId":1,"code":"WXF XWAR C.SHRS
PLN","mic":"XWAR","nominalValue":0,"nominalValueType":"Constant","multipli
er":0,"strikePrice":0,"productIdentification":"AT0000827209","productIdent
ificationType":"ISIN","settlementCalendarId":2,"bondCouponType":"NotApplic
able","preTradeCheckMinPrice":1000000,"preTradeCheckMaxPrice":500000000000
0,"preTradeCheckMinQuantity":1,"preTradeCheckMaxQuantity":2700000,"preTrad
eCheckMaxValue":1000000000,"preTradeCheckMinValue":0,"liquidity":false,"ma
rketModelType":"CLOB","issuer":"Warimpex Finanz- und Beteiligungs
AG","issuerRegCountry":"AUT","underlyingInstrumentId":0,"productCode":"WXF
","cfi":"ESVUFB","fisn":"WARIMPEX/SHS","issueSize":54000000,"nominalCurren
cy":"PLN","usIndicator":"NotApplicable","expiryDate":0,"versionNumber":0,"
settlementType":"NotApplicable","optionType":"NotApplicable","excerciseTyp
e":"NA","productName":"WARIMPEX","referencePrice":450000000,"status":"Acti
ve","initialPhaseId":2}
```


Instrument message for a productType=FinancialProductShare (ISIN AT0000827209). marketStructureId=10 indicates that the Instrument belongs to the element of the Market Structure with Id=10. The instrument follows calendarId=1, the tick table is identified with tickTableId=1, the collar group is collarGroupId=3 and the trading schedule followed by this instrument is tradingScheduleId=1.

7.1.4.3. Tick Tables

TickTableEntry messages combine to create Tick Tables. A Tick Table is a set of price intervals with related steps in which prices can increase.

A tick table:

- must include the price 0 in its first interval (lowerBound),
- does not specify an upper bound in the highest interval,
- specifies that prices are ≥ lowerBound within each interval, but < lowerBound in following interval
- is currency agnostic.


```
{"header":{"length":59,"msgType":"TickTableEntry","version":4096,"seqNum":
41,"timestamp":1705460401056076979,"sourceTimestamp":1705460401056003077,"
isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":50
000,"lowerBound":0,"tickTableId":1}
```


Tick Table 1 makes prices increase in the interval from 0 to ∞ in steps of 10000.

A series of TickTableEntry messages with an identical tickTableId define a tick table.

```
{"header":{"length":59,"msgType":"TickTableEntry","version":4096,"seqNum":160,"timestamp":1705460401072863623,"sourceTimestamp":1705460401072837522,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":1000000,"lowerBound":0,"tickTableId":11}
{"header":{"length":59,"msgType":"TickTableEntry","version":4096,"seqNum":161,"timestamp":1705460401072993726,"sourceTimestamp":1705460401072961225,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":2000000,"lowerBound":1000000000,"tickTableId":11}
{"header":{"length":59,"msgType":"TickTableEntry","version":4096,"seqNum":162,"timestamp":1705460401073111129,"sourceTimestamp":1705460401073084328,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":5000000,"lowerBound":2000000000,"tickTableId":11}
{"header":{"length":59,"msgType":"TickTableEntry","version":4096,"seqNum":163,"timestamp":1705460401073259333,"sourceTimestamp":1705460401073231032,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":10000000,"lowerBound":5000000000,"tickTableId":11}
{"header":{"length":59,"msgType":"TickTableEntry","version":4096,"seqNum":164,"timestamp":1705460401073380436,"sourceTimestamp":1705460401073351235,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":20000000,"lowerBound":10000000000,"tickTableId":11}
```


A series of TickTableEntry messages with tickTableId=11 combine to produce a complete tick table shown below.

Tick table identified by tickTableId=8 starts at the price 0.

Sequence number	Price interval		Tick size
	From	To	
160	0	1000000000	1000000
161	1000000000	2000000000	2000000
162	2000000000	5000000000	5000000
163	5000000000	10000000000	10000000
164	10000000000	∞	20000000

7.2. Trading Calendar Messages

Trading calendar messages define a trading calendar and trading schedule for Instruments (InstrumentID). A Trading Schedule (tradingScheduleId) is a collection of trading phases, each identified by tradingPhaseId. The TradingPhaseScheduleEntry message informs which TradingPhases (identified by tradingPhaseId) are attached to a particular Schedule (tradingScheduleId).

Instrument (InstrumentID) follows a Trading Schedule (tradingScheduleId) and switches between TradingPhases. Instrument tracks a predefined WeekPlan and applies the tradingScheduleId on valid trading days, except for CalendarException days.

The job of an InstrumentStatusChange message is to report that an Instrument changed phases.

7.2.1. TRADINGPHASESCHEDULEENTRY

Trading schedules are instrument agnostic. They are a collection of phases that an instrument will follow. An Instrument will refer to a particular tradingScheduleId.

TradingPhaseScheduleEntry defines trading phases (tradingPhaseId) for a trading schedule. TradingPhaseScheduleEntry messages combine to define a complete trading schedule (tradingScheduleId).

TradingPhaseScheduleEntry functions in conjunction with InstrumentStatusChange which signals the beginning of a new trading phase for an instrument (TradingPhaseId).

7.2.2. CALENDAREXCEPTION

Specifies the calendar date for which an exception is scheduled.

Four types of exceptions can be specified: market Open, Closed, half day and technical exception. The exception can be recurrent or one-time.

Refers to the CalendarID of the corresponding WeekPlan message.

7.2.3. WEEKPLAN

WeekPlan defines the days of the week (Monday-Sunday) during which trading is permitted.

A customized week plan can be devised for each Instrument.

The WeekPlan is assigned a particular CalendarID.

7.2.4. EXAMPLES

Calendar messages are

- defined in abstraction – independently of Markets or instruments. They can be assigned to different Markets or instruments.
- linked by the calendarId identifier or key

7.2.4.1. Week Plans

WeekPlan specifies the days of the week during which trading can occur. A calendarID may only have one week-plan.


```
{"header":{"length":50,"msgType":"WeekPlan","version":4096,"seqNum":3,"timestamp":1705460401053556412,"sourceTimestamp":1705460401052783892,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"calendarId":1,"weekdays":[true,true,true,true,true,false,false]}
```


A week plan with ID 1 (calendarId=1) defines that the trading week lasts from Monday until Friday, and that the market is closed on Saturday and Sunday.

7.2.4.2. Calendar Exceptions

CalendarException is a means of defining exceptions to the regular trading calendar. It provides a method to define calendar days during which trading will not occur on standard terms. A calendar exception is a calendar date on which the market may be closed or opened, trading hours may be reduced (half-day), or where a technical exception may take place.

Market Data will usually communicate a series of exception days for a particular CalendarID. In the example below, a series of messages with calendarId=1 combine with a WeekPlan message similarly identified with calendarId=1. CalendarException assigns exceptions to the week plan identified by the same calendarId.


```
{"header":{"length":49,"msgType":"CalendarException","version":4096,"seqNum":9,"timestamp":1705460401054538438,"sourceTimestamp":1705460401053913822,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"calendarId":1,"calendarExceptionDate":20231101,"calendarExceptionRecurrent":false,"calendarExceptionType":"Closed"}
```


Together, the WeekPlan and CalendarException messages identified by calendarId=1 define that in that particular calendar trading will occur from Monday to Friday, except for November 1st 2023. This date is not recurrent.

7.2.4.3. Trading Schedules and Phases

A Trading Schedule is a collection of Trading Phases. A series of TradingPhaseScheduleEntry messages define trading phases (tradingPhaseId) and assign them to schedules (tradingScheduleId). A series of TradingPhaseScheduleEntry messages with the same tradingScheduleId define a Trading Schedule. An Instrument refers to a Trading Schedule (tradingScheduleId) and follows it during the trading day. Phase changes in a schedule are signaled on the market data feed by the InstrumentStatusChange message. Trading Schedules are Instrument agnostic and can be attached to one or more Financial Products.

For example, the following message stipulates that Trading Schedule 1 only contains one trading phase (tradingPhaseId=1).


```
{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":4096,"seqNum":232,"timestamp":1705460401082165168,"sourceTimestamp":1705460401082138168,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tradingScheduleId":1,"tradingPhaseId":7,"tradingPhaseSettlementCycle":2,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":10,"dynamicCollarVolatilityAuctionId":27,"tradingPhaseStartTime":1705507200000000000,"tradingPhaseStopTime":1705507500000000000,"tradingPhaseType":"ContinuousLastAuctionTime","auctionType":"NotApplicable","auctionUncrossing":false,"extStaticCollarVolatilityAuctionId":11,"extDynamicCollarVolatilityAuctionId":28}
```


In this message, Trading Phase 7 starts at 16:00H UTC and ends at 16:05H UTC on 17 January 2024. The phase is reserved of type ContinuousLastAuctionTime.

The example below shows how a Trading Schedule can be assembled from a series of Trading Phase definitions.


```
{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":4096,"seqNum":247,"timestamp":1705460401084199122,"sourceTimestamp":1705460401084155521,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tradingScheduleId":4,"tradingPhaseId":22,"tradingPhaseSettlementCycle":0,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":1705446000000000000,"tradingPhaseStopTime":1705446300000000000,"tradingPhaseType":"NoTradingClosed","auctionType":"NotApplicable","auctionUncrossing":false,"ext
```

```
StaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0
}
{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":409
6,"seqNum":248,"timestamp":1705460401084318825,"sourceTimestamp":170546040
1084277424,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"
tradingScheduleId":4,"tradingPhaseId":23,"tradingPhaseSettlementCycle":0,"
tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0
,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":170544630000
0000000,"tradingPhaseStopTime":1705446600000000000,"tradingPhaseType":"NoT
radingEarlyMonitoring","auctionType":"NotApplicable","auctionUncrossing":f
alse,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuc
tionId":0}
{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":409
6,"seqNum":249,"timestamp":1705460401084442729,"sourceTimestamp":170546040
1084401128,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"
tradingScheduleId":4,"tradingPhaseId":24,"tradingPhaseSettlementCycle":0,"
tradingPhaseMaxBlockSettlementCycle":30,"staticCollarVolatilityAuctionId":
0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":17054466000
00000000,"tradingPhaseStopTime":1705531800000000000,"tradingPhaseType":"Bl
ockExactMatching","auctionType":"NotApplicable","auctionUncrossing":false,
"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionI
d":0}
{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":409
6,"seqNum":250,"timestamp":1705460401084569432,"sourceTimestamp":170546040
1084542531,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"
tradingScheduleId":4,"tradingPhaseId":25,"tradingPhaseSettlementCycle":0,"
tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0
,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":170553180000
0000000,"tradingPhaseStopTime":1705532100000000000,"tradingPhaseType":"NoT
radingLateMonitoring","auctionType":"NotApplicable","auctionUncrossing":fa
lse,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuct
ionId":0}
{"header":{"length":84,"msgType":"TradingPhaseScheduleEntry","version":409
6,"seqNum":251,"timestamp":1705460401084694735,"sourceTimestamp":170546040
1084669135,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"
tradingScheduleId":4,"tradingPhaseId":26,"tradingPhaseSettlementCycle":0,"
tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0
,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":170553210000
0000000,"tradingPhaseStopTime":1705532340000000000,"tradingPhaseType":"NoT
radingClosed","auctionType":"NotApplicable","auctionUncrossing":false,"ext
StaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0
}
```


In this example, Trading Schedule 4 (tradingScheduleId=4) is composed of 5 different Trading Phases (tradingPhaseId=22, 23, 24, 25, 26).

7.3. Order And Trade Collars

7.3.1. COLLARTABLEENTRY

CollarTableEntry defines Order and Trade Collar (collarMode) by providing the range of deviation from reference prices in percentage form or absolute values (collarExpression).

This message only contains the collar ranges, but not the values of the collars. The values of the collars are communicated by the OrderCollars and TradeCollars messages at the beginning of each trading phase.

7.3.2. COLLARGROUP

A group of collars which includes Static and Dynamic Order collars and Static and Dynamic Trade collars.

7.3.3. EXAMPLES

Market Data communicates the Order and Trade price collars configured by the Market Operator. The key to understanding the difference between the collar types is as follows:

- Order Collars screen all orders submitted to the market,
- Trade Collars guard against large volatility swings in security prices.

While different in nature, Order and Trade collars, together with their parameters, are communicated by CollarTableEntry Market Data messages. To accurately frame an Instrument's price behavior, collars are bundled together into groups with the CollarGroup message. The table below provides a short description of various collars and their properties.

Collar Type	Description	Possible Mode	Mode Description
Order Price Collars	Prevent orders with excessive bid or ask prices from entering the Order Book.	Static	Wide collar. Centered around reference price at beginning of day.
		Dynamic	Narrow collar. Follows the market price in real time.
Trade Price Collars	Ensure that trades in the Order Book occur within reasonable price ranges.	Static	Wide collar. Centered around reference price at beginning of day.
		Dynamic	Narrow collar. Follows the market price in real time.

Collars are always expressed as deviations (in percentage form or absolute value) from a reference price or the real time market price. The collar definition message CollarTableEntry only contains relative deviations and not collar values which may be directly applicable for business purposes. The Market Data messages OrderCollars and TradeCollars convey an Instrument's calculated collar values applicable at any time during the trading session. These values are computed from collar definitions contained in CollarTableEntry and relevant reference prices in effect at the time.

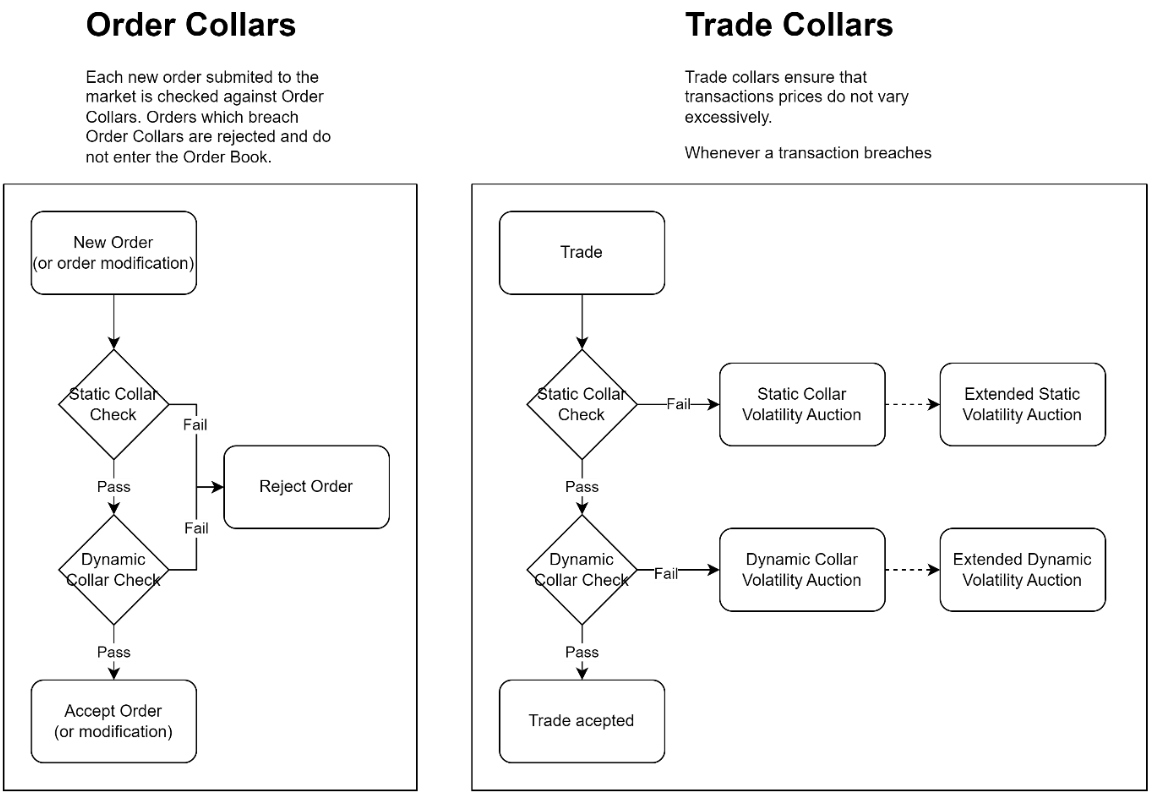
NOTE: because collars are defined through deviations, the same collars assigned to two Instruments will obtain different values if the reference prices for those Instruments differ.

To summarize, Trade and Order price collars are set and configured by two Market Data messages, CollarGroup and CollarTableEntry:

- CollarTableEntry defines collars,
- CollarGroup groups the collars in a single structure used to control trading behavior,

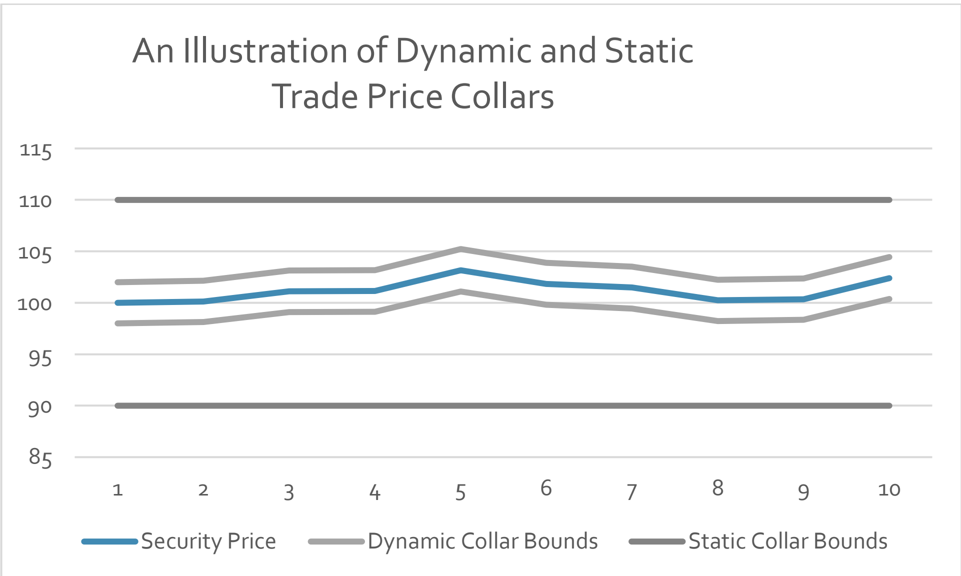
- OrderCollars and TradeCollars convey the calculated collar values for the relevant Instrument.

Figure 6 - Collar Mechanisms



In the figure chart, a Dynamic Trade Price Collar of 2% and a Static Trade Price Collar of 10% around the Reference Price are shown.

Figure 7 - Static and Dynamic Collars



7.3.3.1. Collar Groups

A Collar Group (CollarGroup) binds a set of 4 Collar Tables to produce a complete definition of collars. A Collar Group consists of 4 collar tables specified and defined by CollarTableEntry messages:

- Static collars for Orders
- Dynamic collars for Orders
- Static collars for Trades
- Dynamic collars for Trades

To fully configure a collar group, a set of 4 Collar Tables must be specified. However, depending on the application, a collar group may contain fewer than 4 collar tables. For example, one Collar Group may only feature Trade Collars, another only Order Collars while still another may only contain Order Static Collars. The choice is driven by regulations, Market rules and the definition of Instruments.

Multiple CollarTableEntry messages may be needed to specify a single Collar Table. In such a case Market Data will broadcast several CollarTableEntry messages with the same collarTableId.

Static and Dynamic collars, for both Orders and Trades, are defined in the same manner but must be encoded in different tables. For example, one CollarTableEntry table needs to be defined for a Static Trade collar and another for a Dynamic Trade collar.

Instruments refer to specified Collar Groups. A single Collar Group can be attached to one or more Instruments and assign the same set of collars to these instruments.

The CollarGroup message only requires the identifiers to relevant CollarTableEntry definitions. These fields are shown below:

Field name	Description	Relevant values from the example below
id	Collar Group identifier.	17
orderStaticCollarTableId	ID of Collar Table.	21
orderDynamicCollarTableId	ID of Collar Table.	22
tradeStaticCollarTableId	ID of Collar Table.	31
tradeDynamicCollarTableId	ID of Collar Table.	32

{"header":{"length":59,"msgType":"CollarGroup","version":4096,"seqNum":3528,"timestamp":1705460401717294356,"sourceTimestamp":1705460401717226254,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"id":10,"orderStaticCollarTableId":27,"orderDynamicCollarTableId":0,"tradeStaticCollarTableId":23,"tradeDynamicCollarTableId":10}

This Collar Group binds Collar Tables 27, 23, 10 into a complete collar definition with id=10.

7.3.3.2. Collar Tables

CollarTableEntry messages define individual Collar Tables. Those consist of a definition of deviations from a reference price. Collar tables are independent definitions not directly dependent on Instruments which they are meant to service. Each collar table is identified by its own collarTableId.

The table below explains the relevant fields in the CollarTableEntry message and examples below illustrate simple examples of collar definitions.

Field name	Description
collarTableId	The identifier of the Collar Table. Referred to in CollarGroup.
collarMode	OrderPriceCollar for Order Collars or TradePriceCollar for Trade Collars.
collarExpression	Either AbsoluteValue or Percentage. • If AbsoluteValue the fields collarValue, collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk contain absolute currency values (e.g., USD). • If Percentage, the fields collarValue, collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk contain percentages.
collarLowerBound	The lower bound of the reference price for which the collar defined in the message is applicable. Several CollarTableEntry may be used to define a "ladder" of collars. If collarLowerBound=0, the collar is applicable in the reference price interval from 0 to ∞, or the collarLowerBound of the following table.
collarValue	Field applicable to Trade price collars (collarMode=TradePriceCollar). Contains the allowed deviation of Trade prices from a reference price. Deviations expressed in absolute values or percentages. The format depends on collarExpression={AbsoluteValue, Percentage}. 0 for Order price collar (collarMode=OrderPriceCollar).
collarLowerBid **collarUpperBid** **collarUpperAsk** **collarLowerAsk**	Fields applicable to Order price collars only (collarMode=OrderPriceCollar) contain limits on Bids and Asks for incoming orders. Deviations expressed in absolute values or percentages. The format depends on collarExpression={AbsoluteValue, Percentage}. 0 for Trade price collar (collarMode=TradePriceCollar).

The table below illustrates collar calculations that are valid for each rung of the ladder defined by collarLowerBound and for Static and Dynamic collars. The calculations depend on the Reference Price (RefPrice) in force for a given Instrument. It is important to note that CollarTableEntry only defines collars in terms of deviations. OrderCollars and TradeCollars convey calculated collar ranges.

Collar Type	Computation of collar ranges for a given Reference Price
Order Price Collar	If collar defined in percentages (collarExpression=Percentage) RefPrice*(1 - collarLowerBid) ≤ Buy OrderPrice ≤ RefPrice*(1 + collarUpperBid) RefPrice*(1 - collarLowerAsk) ≤ Sell OrderPrice ≤ RefPrice*(1 + collarUpperAsk) If collar defined in absolute values (collarExpression=AbsoluteValue) RefPrice - collarLowerBid ≤ Buy OrderPrice ≤ RefPrice + collarUpperBid RefPrice - collarLowerAsk ≤ Sell OrderPrice ≤ RefPrice + collarUpperAsk
Trade Price Collar	If collar defined in percentages (collarExpression=Percentage) RefPrice*(1 - collarValue) ≤ Trade Price ≤ RefPrice*(1 + collarValue) If collar defined in absolute values (collarExpression=AbsoluteValue) RefPrice - collarValue ≤ Trade Price ≤ RefPrice + collarValue


```
{"header":{"length":93,"msgType":"CollarTableEntry","version":4096,"seqNum":166,"timestamp":1705460401073574941,"sourceTimestamp":1705460401073553641,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"collarTableId":1,"collarMode":"TradePriceCollar","collarExpression":"Percentage","collarLowerBound":0,"collarValue":600000000,"collarLowerBid":0,"collarUpperBid":0,"collarUpperAsk":0,"collarLowerAsk":0}
{"header":{"length":93,"msgType":"CollarTableEntry","version":4096,"seqNum":167,"timestamp":1705460401073726245,"sourceTimestamp":1705460401073699345,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"collarTableId":1,"collarMode":"TradePriceCollar","collarExpression":"Percentage","collarLowerBound":20000000,"collarValue":300000000,"collarLowerBid":0,"collarUpperBid":0,"collarUpperAsk":0,"collarLowerAsk":0}
```

```
{"header":{"length":93,"msgType":"CollarTableEntry","version":4096,"seqNum":221,"timestamp":1705460401080763831,"sourceTimestamp":1705460401080726630,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"collarTableId":30,"collarMode":"OrderPriceCollar","collarExpression":"AbsoluteValue","collarLowerBound":1000000,"collarValue":0,"collarLowerBid":500000000,"collarUpperBid":600000000,"collarUpperAsk":400000000,"collarLowerAsk":300000000}
{"header":{"length":93,"msgType":"CollarTableEntry","version":4096,"seqNum":222,"timestamp":1705460401080879634,"sourceTimestamp":1705460401080847834,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"collarTableId":30,"collarMode":"OrderPriceCollar","collarExpression":"Percentage","collarLowerBound":10000000000,"collarValue":0,"collarLowerBid":900000000,"collarUpperBid":1000000000,"collarUpperAsk":800000000,"collarLowerAsk":700000000}
```


An example of 2 Collar Tables. collarTableId=1 defines an Trade price collar in terms of percentages while collarTableId=30 defines an Order price collar in absolutes and percentages of reference price.

Let's decode these messages. The table below highlights the differences between the structure of Order and Trade collar definitions.

Field name	Message 221 Order Collar (`seqNum=221`)	Message 166 Trade Collar (`seqNum=166`)
collarTableId	30	1
collarMode	OrderPriceCollar	TradePriceCollar
collarExpression	AbsoluteValue	Percentage
collarLowerBound	1000000	0
collarValue	0	600000000
collarLowerBid	500000000	0
collarUpperBid	600000000	0
collarUpperAsk	400000000	0
collarLowerAsk	300000000	0

collarTableId=30 is an Order price collar expressed in Absolute Value. collarLowerBound=0 means the collar definition is applicable in the price interval from 1000000 to ∞. As this is an Order Price Collar, the field collarValue=0 (this is always the case for Order Price Collars). The collar absolute deviations are conveyed by the fields collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk. These are absolute deviations from the reference price and represent currency values. The deviations are 50₵, 60₵, 40₵, and 30₵ from the relevant reference price.

collarTableId=1 is a Trade price collar expressed in Percentages. collarLowerBound=0 means the collar definition is applicable in the price interval from 0 to ∞. The collar deviation is 60% defined in the field collarValue=60%. As this is a Trade Price Collar the fields collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk are left undefined and are set to 0.

7.3.4. FURTHER EXAMPLES

As the nature of collars is complex, we include some further examples to clarify the concepts involved.

7.3.4.1. Percentage vs Absolute Value Expressions

This table shows collar definitions in Absolute Values and Percentages. The only difference is the collarExpression field and the change in value types in relevant message fields.

Field name	Order Collar		Trade Collar	
	Percentages	Absolute Value	Percentages	Absolute Value
collarTableId	17	18	23	24
collarMode	OrderPriceCollar	OrderPriceCollar	TradePriceCollar	TradePriceCollar
collarExpression	Percentage	AbsoluteValue	Percentage	AbsoluteValue
collarLowerBound	0	0	0	0
collarValue	0	0	10%	₵100
collarLowerBid	5%	₵50	0	0
collarUpperBid	6%	₵60	0	0
collarUpperAsk	4%	₵40	0	0
collarLowerAsk	3%	₵30	0	0

7.3.4.2. Collar Ladders

Collars can be layered in the same manner as tick tables. Consequently, collar ranges can vary in relation to the value of the reference price. For example, when the reference price is low, collars can be defined to be narrow and broad when the reference price increases. This capability is enabled by the collarTableId and collarLowerBound fields in CollarTableEntry. Collar ladders can be applied to order and trade collars (via collarMode: OrderPriceCollar or TradePriceCollar).

Collar ladders must start at 0 – this is expressed by collarLowerBound=0. In the example below, to define a 3-level collar, 3 CollarTableEntry messages with the same collarTableId are broadcast. They differ in the collarLowerBound value and collar parameters.


```
{"header":{"length":93,"version":4096,"msgType":"CollarTableEntry","seqNum
":208,"timestamp":"2023-02-10T02:32:23.549452Z","sourceTimestamp":"2023-
02-
10T02:32:23.547959Z","isEncrypted":false,"encryptionOffset":0,"encryptionK
eyId":0},"collarTableId":18,"collarMode":"OrderPriceCollar","collarExpress
ion":"Percentage","collarLowerBound":0,"collarValue":0,"collarLowerBid":90
0000000,"collarUpperBid":1000000000,"collarUpperAsk":800000000,"collarLowe
rAsk":700000000}
{"header":{"length":93,"version":4096,"msgType":"CollarTableEntry","seqNum
":209,"timestamp":"2023-02-10T02:32:23.549373Z","sourceTimestamp":"2023-
02-
10T02:32:23.547573Z","isEncrypted":false,"encryptionOffset":0,"encryptionK
eyId":0},"collarTableId":18,"collarMode":"OrderPriceCollar","collarExpress
ion":"AbsoluteValue","collarLowerBound":1000000,"collarValue":0,"collarLow
erBid":500000000,"collarUpperBid":600000000,"collarUpperAsk":400000000,"co
llarLowerAsk":300000000}
{"header":{"length":93,"version":4096,"msgType":"CollarTableEntry","seqNum
":210,"timestamp":"2023-02-10T02:32:23.549452Z","sourceTimestamp":"2023-
02-
10T02:32:23.547959Z","isEncrypted":false,"encryptionOffset":0,"encryptionK
eyId":0},"collarTableId":18,"collarMode":"OrderPriceCollar","collarExpress
ion":"Percentage","collarLowerBound":10000000000,"collarValue":0,"collarLo
```

werBid":900000000,"collarUpperBid":1000000000,"collarUpperAsk":800000000,"collarLowerAsk":700000000}

The messages with collarTableId=18 imply that the Collar table has 3 different levels. The 3 levels are specified by collarLowerBound={0, 10, 100}.

Let's break down this series of messages.

Field name	Message 208 (seqNum=208)	Message 209 (seqNum=209)	Message 210 (seqNum=210)
collarTableId	18	18	18
collarMode	OrderPriceCollar	OrderPriceCollar	OrderPriceCollar
collarExpression	Percentage	AbsoluteValue	Percentage
collarLowerBound	$100	$10	$100
collarValue	0	0	0
collarLowerBid	9%	₵50	9%
collarUpperBid	10%	₵60	10%
collarUpperAsk	8%	₵40	8%
collarLowerAsk	7%	₵30	7%

The three messages each refer to collarTableId=18. This signals that they form a collar ladder. They also define Order Price Collars. Message 208 states that collarLowerBound=0, message 209 features collarLowerBound=10 and message 210, collarLowerBound=100. This series of messages defines a 3-level collar. The final spread of the collars clearly depends on the value of the reference price - when the price breaks through $10 or $100, the range of the collars changes. The calculated collar values in currency units are communicated by OrderCollars and TradeCollars as the reference price varies relative to the collarLowerBounds defined.

Collar level	Value of Reference Price	Collar Calculation reported by **OrderCollars and TradeCollars**
1	0 < RefPrice < $10	Collar is defined in percentages (collarExpression=Percentage) RefPrice*(1 - 9%) ≤ Buy OrderPrice ≤ RefPrice*(1 + 10%) RefPrice*(1 - 7%) ≤ Sell OrderPrice ≤ RefPrice*(1 + 8%)
2	$10 ≤ RefPrice < $100	Collar is defined in absolute values (collarExpression=AbsoluteValue) RefPrice - ₵50≤ Buy OrderPrice ≤ RefPrice + ₵60 RefPrice - ₵30 ≤ Sell OrderPrice ≤ RefPrice + ₵40
3	$100 ≤ RefPrice < ∞	Collar is defined in percentages (collarExpression=Percentage) RefPrice*(1 - 9%) ≤ Buy OrderPrice ≤ RefPrice*(1 + 10%) RefPrice*(1 - 7%) ≤ Sell OrderPrice ≤ RefPrice*(1 + 8%)

7.4. Pricing Messages

7.4.1. ACCRUEDINTERESTTABLEENTRY

AccruedInterestTableEntry shows the accrued interest for a financial product (ProductID). A financial product is the archetype of an instrument (identified by InstrumentID). An instrument (InstrumentID) refers to a financial product (ProductID) in its Instrument definition message.

{"header":{"length":55,"msgType":"AccruedInterestTableEntry","version":4096,"seqNum":358,"timestamp":1705460401107991651,"sourceTimestamp":1705460401097592876,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"productId":949,"accruedInterestTableEntryDate":20240117,"accruedInterestTableEntryValue":364800}

A sample AccruedInterestTableEntry message.

7.4.2. INDEXATIONTABLEENTRY

IndexationTableEntry behaves similarly to AccruedInterestTableEntry. It shows a bond instrument's indexed nominal value. An instrument (InstrumentID) refers to a financial product (ProductID) in its Instrument definition message.

8. Index Messages

The Index GPW WATS facility conveys the value of market indices throughout the trading day together with end-of-day statistics.

The order of Index messages during a trading day is as follows:

Index Message	Message Type	Note
IndexParams	Parametrization	The first Index message sent during parametrization. It frames the index.
IndexPortfolioEntry	Parametrization	Sequence of messages following an IndexParams message. Lists the Instruments which compose of the index.
RealTimeIndex	Realtime data	The value of the index broadcast during the trading day.
IndexSummary	End of day data	Index statistics at session close.

8.1. IndexParams

Carries metadata related to an index. This information contains parameters such as the base value of the index, the number of components and currency.

8.2. IndexPortfolioEntry

Defines an index by reference to the instruments traded in the GPW WATS platform. The definition of the index is made by giving a list of InstrumentId relating to specific instruments. The index is defined using a series of IndexPortfolioEntry messages, one for each security composing the index. For example, in order to specify an index composed of 30 securities, 30 IndexPortfolioEntry messages will be broadcast on Market Data.

8.3. RealTimeIndex

Communicates the real time value of the index and related statistics.

Index information is sent at intervals appropriate for each respective market index and the publication intervals are parametrized at system level. A variety of publication intervals can be pre-configured ranging from near-real time publication (15 seconds or less) to several times per day.

The publication frequency is dependent on the specification of the index.

8.4. IndexSummary

This is an end-of-trading-day message. It provides the end-of-day value of the index and summarizes the variation in the index during the trading day with select statistics including high, low and closing value.

9. Session Management Messages

9.1. StartOfTechnicalSession

The first message sent by the Stream service during the technical session which marks the start of that session.


```
{"header":{"length":41,"msgType":"StartOfTechnicalSession","version":4096,"seqNum":1,"timestamp":1705431616701236647,"sourceTimestamp":1705431616701139244,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"sessionId":600}
```


9.2. EndOfTechnicalSession

The last message sent by the Stream service during the technical session marks the end of that session.

9.3. ReferenceDataStart

ReferenceDataStart is used by the Snapshot and Stream services.

It marks the start of the series of messages which contain the reference data.


```
{"header":{"length":39,"msgType":"ReferenceDataStart","version":4096,"seqNum":2,"timestamp":1705460401053487310,"sourceTimestamp":1705460401047443551,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0}}
```


9.4. ReferenceDataEnd

ReferenceDataEnd is used by the Snapshot and Stream services and marks the end of the series of messages which contain the reference data.


```
{"header":{"length":39,"msgType":"ReferenceDataEnd","version":4096,"seqNum":15667,"timestamp":1705460402649234289,"sourceTimestamp":1705460402649203988,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0}}
```


9.5. EndOfSnapshot

This message sent by the Snapshot service indicates the end of the snapshot.

Following the message, the Snapshot service disconnects automatically.

EndOfSnapshot contains the last sequence number – lastSeqNum.

This number is used to synchronize with the messages from the real time Stream service. Under optimal conditions, after retrieving the Snapshot, the user should capture the message numbered lastSeqNum+1 on Stream. The Replay service can be accessed any messages are dropped.

9.6. EncryptionKey

The encryption keys used to decrypt the relevant Market Data messages are disseminated daily by the Snapshot service.

Each Key is accompanied by a label (key ID) which identifies the key. Messages will refer to particular KeyIDs to be decrypted.

9.7. ReplayRequest

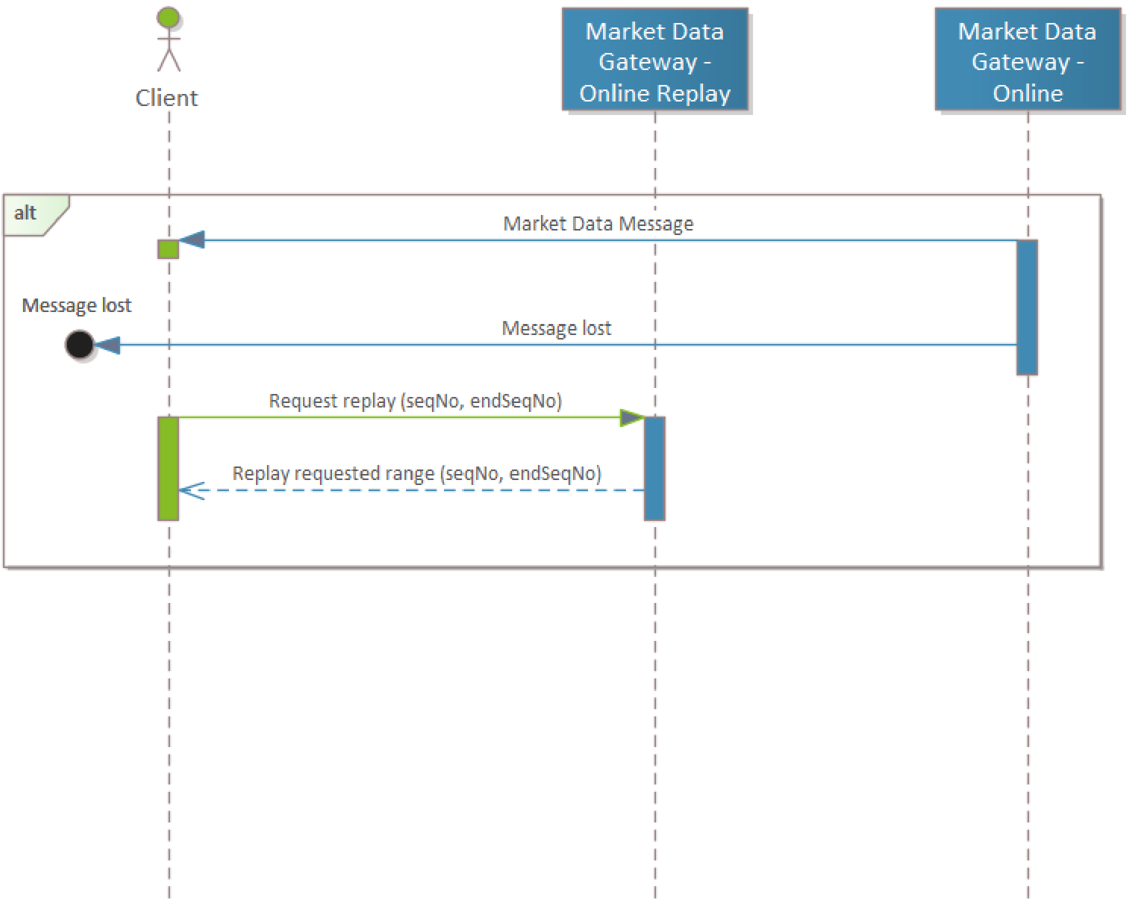
The ReplayRequest message supports the single use case of recovering missing messages from Stream. In this situation ReplayRequest repeats messages referenced by a range of sequence numbers.

A replay request contains two parameters: seqNum (starting sequence number) and endSeqNum (ending sequence number):

- If seqNum==0, the replay starts from the first message.
- If endSeqNum==0, the replay ends on the last message.

The Replay service sets an upper bound on the number of messages that can be replayed in one request: (endSeqNo -seqNo) < SystemMaximum. The maximum amount is defined by the System Operator. When there is a large message gap or when a Participant joins the trading platform late during the market session, a Snapshot should be used.

Figure 8 - OMD Replay Behavior



10. General Considerations

10.1. GPW WATS Time

GPW WATS runs on UTC time and GPW WATS Market Data serves information signed by UTC time stamps. All messages and all fields which include a time stamp are to be assumed UTC. The timestamp and sourceTimestamp fields in the Header of Market Data messages represent UTC time.

10.2. Matching Engine Failure

Under normal operating conditions, the matching engine sends real time messages for each market data stream from redundant sites A and B. Users may implement line arbitration mechanisms, but this is not recommended because of inter-site communication delays. Replay should be used instead.

10.3. Price-Time Priority Determination

Each message sent on Market Data features a Header. The Header contains the SeqNum field which represents the sequence number assigned to an order by the Market Data Sequencer. The priority of an order should be determined using this value. For example, for orders at the same price level the order with the lower SeqNum has priority over an order with a higher SeqNum.

One should note that changes to order priority following order modification are communicated in the OrderModify message by the PriorityFlag value - an order can either retain or lose priority following modification.

10.4. Encryption

Market Data is secured to prevent unauthorized access to data. Encryption is performed by the Encryption Engine component and involves encrypting sensitive data in online Market Data messages. The list of fields in messages subject to encryption is fixed. Encryption involves the use of a key which is generated per session and only valid for that session.

Encryption keys are disseminated by the Snapshot service before the start of each trading day by EncryptionKey messages. The encryptionKeyId is used to identify the appropriate encryption keys for different messages.

GPW WATS implements the ChaCha20 encryption algorithm in CTR (i.e., counter) mode. In this mode, the following chacha20 initialization should be performed:

- Encryption Key - the Online Market Data Header message refers to an encryption key (encryptionKeyId) which is distributed by EncryptionKey messages, as a part of Market Data snapshot messages.
- Nonce - The value for the nonce is not secret and is assumed to be the session date ASCII byte string YYYYMMDD, left padded with zeros to form a proper 96 bitfield. It is an unsigned 8-bit integer.
- The counter starts with the value 0 and is encoded as a 32 bit integer.

A provisioned stream cipher object is created for each captured key and is used to decrypt designated fields of messages. In most implementations there is no need to alter any of the initial parameters of the encryption object, such as counter, as they should be properly set by the implementation, which provides keystream for decryption as the stream of data progresses. Each Header message contains an encryptionOffset which determines the absolute offset in the encryption stream used to encrypt the message. The decryption process should only be performed if the isEncrypted field in the Header message is true.

The list of encrypted Market Data messages is as follows:

Message types	Message	Encrypted Fields	Note
Continuous Trading	OrderAdd	InstrumentId price quantity	
	OrderModify	InstrumentId price quantity	
BBO	PriceLevelSnapshot	price quantity orderCount	In this message instrumentid is not encrypted for technical reasons.
Execution and Trade	OrderExecute	quantity InstrumentId executionId executionPrice executionQuantity	

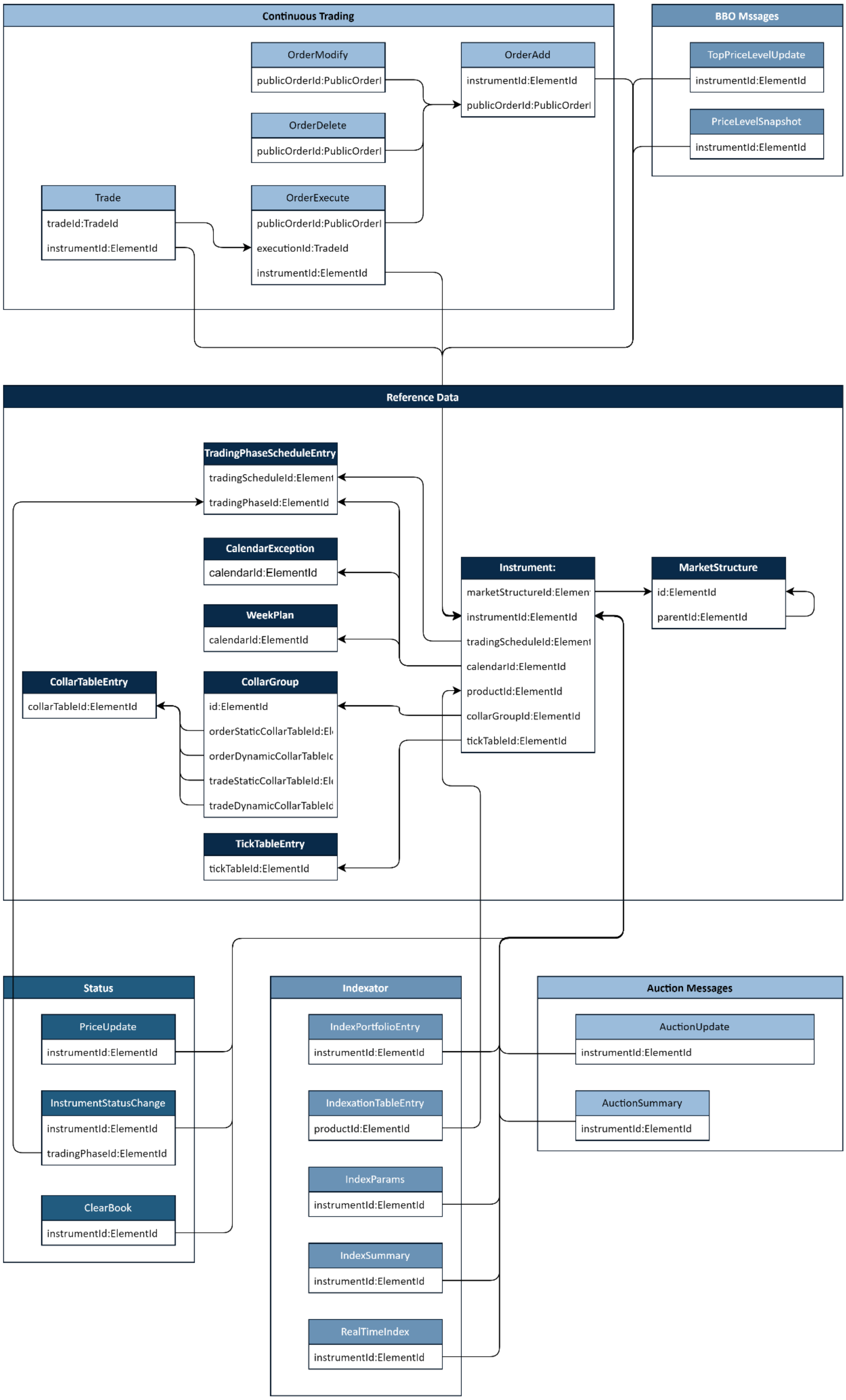
10.5. Market Data Suppression

Market Data does not broadcast all information related to trading activity and suppresses selected information. The data suppression scheme does not hide relevant information from market participants but instead serves to streamline and enhance the performance of the Market Data service.

Order type	Note
Market	Market orders are suppressed and are not communicated by Market Data since they do not persist in the order book. Market Data broadcasts Trade and OrderExecute messages related to Market orders.
Limit	Limit orders, unexecuted or partially executed, are disclosed on Market Data.
Iceberg	Iceberg peaks are displayed since they function as limits.
Stop	The behavior of Stop-orders on Market Data varies. In general, Stop-orders are not published on Market Data until they activate and enter the Order Book. Stop-orders which activate into market orders follow rules for Market orders and are suppressed on Market Data. Stop-orders which activate into limit orders follow rules for Limits and are published on Market Data.

10.6. Market Data Interconnect Diagram

Figure 9 - Market Data Messages



11. Market Data Protocol

11.1. List of Market Data Messages

Name	Data type	Kind	Length	Description
AccruedInterestTableEntry	AccruedInterestTableEntry	Message	55	Message describing an accrued interest table entry.
AuctionSummary	AuctionSummary	Message	60	Auction summary, including auction price and quantity.
AuctionUpdate	AuctionUpdate	Message	107	Updates IMP/IMV and BBO during an auction.
CalendarException	CalendarException	Message	48	Upcoming calendar day exception.
CollarGroup	CollarGroup	Message	59	Message defining a collar group.
CollarTableEntry	CollarTableEntry	Message	93	Collar table definition.
EncryptionKey	EncryptionKey	Message	75	Encryption key and ID.
EndOfSnapshot	EndOfSnapshot	Message	43	Marks the end of a snapshot.
EndOfTechnicalSession	EndOfTechnicalSession	Message	41	End of a technical session.
Heartbeat	Heartbeat	Message	39	A message type used to check connectivity.
IndexParams	IndexParams	Message	179	The metadata message for the given index.
IndexPortfolioEntry	IndexPortfolioEntry	Message	92	The message contains information about a product providing the index portfolio. The entire index portfolio is sent as successive IndexPortfolioEntry messages.
IndexSummary	IndexSummary	Message	115	The index summary message provides a summary of the data calculated in a stock index for a given day.
IndexationTableEntry	IndexationTableEntry	Message	55	Message describing an indexation table entry.
Instrument	Instrument	Message	557	Definition of a financial instrument.
InstrumentStatusChange	InstrumentStatusChange	Message	49	Start of a new trading phase.
InstrumentSummary	InstrumentSummary	Message	133	Provides brief instrument summary.
Login	Login	Message	49	Login message to the Snapshot service.
LoginResponse	LoginResponse	Message	40	Response to a Login message.
Logout	Logout	Message	39	A Logout message is sent by the client to initiate disconnection from the service.
MarketStructure	MarketStructure	Message	102	Messages defining the Trading Venue's market structure.
News	News	Message	925	News message.

Name	Data type	Kind	Length	Description
OpenPositions	OpenPositions	Message	64	The number of open positions in a financial instrument.
OrderAdd	OrderAdd	Message	68	New (limit) order added to order book.
OrderBookEvent	OrderBookEvent	Message	44	Order book event for given instrument
OrderCollars	OrderCollars	Message	84	New order price collars.
OrderDelete	OrderDelete	Message	51	Order deleted.
OrderExecute	OrderExecute	Message	79	Execution report.
OrderModify	OrderModify	Message	68	Order modified.
PriceLevelSnapshot	PriceLevelSnapshot	Message	664	BBO price levels and volumes.
PriceUpdate	PriceUpdate	Message	52	Message indicating a price update
RealTimeIndex	RealTimeIndex	Message	126	Message providing real-time index values for instruments.
ReferenceDataEnd	ReferenceDataEnd	Message	39	Marks the end of the reference data.
ReferenceDataStart	ReferenceDataStart	Message	39	Marks the start of the reference data.
ReplayRequest	ReplayRequest	Message	47	Request to replay a contiguous segment of sequence numbers.
SessionSummary	SessionSummary	Message	373	Session day instrument summary.
StartOfTechnicalSession	StartOfTechnicalSession	Message	41	Start of a technical session.
TickTableEntry	TickTableEntry	Message	59	Tick size definition.
TopPriceLevelUpdate	TopPriceLevelUpdate	Message	62	BBO Level 1.
Trade	Trade	Message	153	Enriched trade report.
TradeCollars	TradeCollars	Message	68	New trade price collars.
TradingPhaseScheduleEntry	TradingPhaseScheduleEntry	Message	80	Trading phase definition.
WeekPlan	WeekPlan	Message	50	Trading week plan.

11.2. MARKET DATA MESSAGE SPECIFICATIONS

11.2.1. HEADER:

Name	Data type	Kind	Length	Description
length	MsgLength	Alias (u16)	2	Message length.
msgType	MsgType	Enum	2	Type of message (e.g. OrderExecute).
version	MsgVersion	Alias (u16)	2	Version of the Market Data protocol.

Name	Data type	Kind	Length	Description
seqNum	SeqNum	Alias (u32)	4	Sequence number of the message added by the Market Data Sequencer.
timestamp	Timestamp	Alias (u64)	8	Timestamp indicating when the message was sequenced.
sourceTimestamp	Timestamp	Alias (u64)	8	Timestamp added by the service which sent the message.
isEncrypted	bool	Primitive	1	True if message is encrypted.
encryptionOffset	u64	Primitive	8	Encryption byte offset.
encryptionKeyId	ElementId	Alias (u32)	4	Reference to an EncryptionKey message.

11.2.2. ACCRUEDINTERESTTABLEENTRY:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
productId	ElementId	Alias (u32)	4	
accruedInterestTableEntryDate	Date	Alias (u32)	4	Accrued interest table entry date.
accruedInterestTableEntryValue	Number	Alias (i64)	8	Accrued interest table entry value.

11.2.3. AUCTIONSUMMARY:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Auction price (final auction price).</td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Auction quantity (final auction volume).</td></tr>
<tr><td>auctionType</td><td>AuctionType</td><td>Enum</td><td>1</td><td>Type of auction.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>NotApplicable.</td></tr>
<tr><td>OpeningAuction</td><td>2</td><td>Opening auction.</td></tr>
<tr><td>ClosingAuction</td><td>3</td><td>Closing auction.</td></tr>
<tr><td>IntradayAuction</td><td>4</td><td>Intraday auction.</td></tr>
<tr><td>VolatilityAuctionStatic</td><td>5</td><td>Volatility auction after static collars breach.</td></tr>
<tr><td>VolatilityAuctionDynamic</td><td>6</td><td>Volatility auction after dynamic collars breach.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table><tr><td>ExtendedVolatilityAuctionStatic</td><td>7</td><td></td></tr><tr><td>ExtendedVolatilityAuctionDynamic</td><td>8</td><td></td></tr><tr><td>UnsuspensionAuction</td><td>9</td><td></td></tr></table></td></tr>
</table>

11.2.4. AUCTIONUPDATE:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
indicativeMatchingPrice	Price	Alias (Number)	8	Indicative Matching Price (IMP).
indicativeMatchingVolume	Quantity	Alias (u64)	8	Indicative Matching Volume (IMV).
totalSellQuantityAtImp	Quantity	Alias (u64)	8	Total sell quantity at IMP.
totalBuyQuantityAtImp	Quantity	Alias (u64)	8	Total buy quantity at IMP.
bestSellLevel	Price	Alias (Number)	8	Best sell price level (1st BBO level).
bestSellLevelQuantity	Quantity	Alias (u64)	8	Quantity at the best sell price level.
bestBuyLevel	Price	Alias (Number)	8	Best buy price level (1st BBO level).
bestBuyLevelQuantity	Quantity	Alias (u64)	8	Quantity at the best buy price level.

11.2.5. CALENDAREXCEPTION:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>calendarId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Calendar ID.</td></tr>
<tr><td>calendarExceptionDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Calendar exception date.</td></tr>
<tr><td>calendarExceptionType</td><td>CalendarExceptionType</td><td>Enum</td><td>1</td><td>Calendar exception type.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Closed</td><td>1</td><td>The day is closed for trading.</td></tr><tr><td>Open</td><td>2</td><td>The day is open for trading.</td></tr></table></td></tr>
</table>

11.2.6. COLLARGROUP:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
id	ElementId	Alias (u32)	4	Collar group ID.
orderStaticCollarTableId	ElementId	Alias (u32)	4	Collar table ID.
orderDynamicCollarTableId	ElementId	Alias (u32)	4	Collar table ID.
tradeStaticCollarTableId	ElementId	Alias (u32)	4	Collar table ID.
tradeDynamicCollarTableId	ElementId	Alias (u32)	4	Collar table ID.

11.2.7. COLLARTABLEENTRY:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>collarTableId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Collar table ID.</td></tr>
<tr><td>collarMode</td><td>CollarMode</td><td>Enum</td><td>1</td><td>Collar mode (trade price collar or order price collar).
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>TradePriceCollar</td><td>1</td><td>Trade price collar.</td></tr>
<tr><td>OrderPriceCollar</td><td>2</td><td>Order price collar.</td></tr>
</table></td></tr>
<tr><td>collarExpression</td><td>ExpressionType</td><td>Enum</td><td>1</td><td>Collar expression (percentage or absolute value in reference price).
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Percentage</td><td>1</td><td>Percentage.</td></tr>
<tr><td>AbsoluteValue</td><td>2</td><td>Absolute value.</td></tr>
</table></td></tr>
<tr><td>collarLowerBound</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Collar lower bound.</td></tr>
<tr><td>collarValue</td><td>CollarValue</td><td>Alias (Number)</td><td>8</td><td>Collar value.</td></tr>
<tr><td>collarLowerBid</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Lower bid collar.</td></tr>
<tr><td>collarUpperBid</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Upper bid collar.</td></tr>
<tr><td>collarUpperAsk</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Upper ask collar.</td></tr>
</table>

Name	Data type	Kind	Length	Description
collarLowerAsk	Bound	Alias (Number)	8	Lower ask collar.

11.2.8. ENCRYPTIONKEY:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
id	ElementId	Alias (u32)	4	Encryption key ID.
secretKey	EncryptionGroupKey	Array (u8)	32	Secret Key value.

11.2.9. ENDOFSNAPSHOT:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
lastSeqNum	SeqNum	Alias (u32)	4	Sequence number of the last Market Data message transmitted by the Snapshot.

11.2.10. ENDOFTECHNICALSESSION:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
sessionId	SessionId	Alias (u16)	2	ID of the session.

11.2.11. HEARTBEAT:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.

11.2.12. INDEXPARAMS:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
indexBaseValue	IndexValue	Alias (Price)	8	Index base value.

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>indexBaseCapitalisation</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Index base capitalisation</td></tr>
<tr><td>indexBaseDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Index base date.</td></tr>
<tr><td>correctionCoefficient</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Index adjustment coefficient K(t)</td></tr>
<tr><td>numOfComponents</td><td>u16</td><td>Primitive</td><td>2</td><td>Total number of instruments in the portfolio.</td></tr>
<tr><td>indexPublicationSchedule</td><td>IndexPublicationSchedule</td><td>Enum</td><td>1</td><td>Mode of index dissemination.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Continuous</td><td>1</td><td>Index is quoted in determined time intervals.</td></tr>
<tr><td>Fixing</td><td>2</td><td>Index is quoted on determined hours.</td></tr>
<tr><td>Closing</td><td>3</td><td>Index is quoted on closing.</td></tr>
</table></td></tr>
<tr><td>indexType</td><td>IndexType</td><td>Enum</td><td>1</td><td>Index product classification
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>W</td><td>1</td><td>WSE index from the main or alternative markets.</td></tr>
<tr><td>Z</td><td>2</td><td>External index.</td></tr>
<tr><td>V</td><td>3</td><td>Another WSE information product.</td></tr>
<tr><td>M</td><td>4</td><td>WSE strategy dividend point index.</td></tr>
<tr><td>S</td><td>5</td><td>WSE strategy index of short type.</td></tr>
<tr><td>L</td><td>6</td><td>WSE strategy index of leverage type.</td></tr>
</table></td></tr>
<tr><td>daysSinceLastPublication</td><td>u8</td><td>Primitive</td><td>1</td><td>Number of calendar days between current session and the previous session day.</td></tr>
<tr><td>numberOfDividends</td><td>u16</td><td>Primitive</td><td>2</td><td>Number of dividends included in the index calculation.</td></tr>
<tr><td>indexUnderlying</td><td>IndexUnderlying</td><td>Array (PublicProductIdentification)</td><td>93</td><td>List of underlying instruments for index.
PublicProductIdentification:
<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>productIdentification</td><td>ProductIdentification</td><td>Array (AnsiChar)</td><td>30</td><td>Product identification, for example its ISIN code.</td></tr>
</table></td></tr>
</table>

Name	Data type	Kind	Length	Description
				productIdentificationType · ProductIdentificationType · Enum · 1 · Type of product identification. Name · Value · Description **ISIN** · 1 · Financial instrument's ISIN
publicationOrder	u16	Primitive	2	Index publication presentation order.
currency	Currency	Enum	2	Currency (e.g. USD).
dateValidity	Date	Alias (u32)	4	Date of validity of index.

11.2.13. INDEXPORTFOLIOENTRY:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
id	ElementId	Alias (u32)	4	The identifier of the entry within the given index.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument.
publicProductIdentification.productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification, for example its ISIN code.
publicProductIdentification.productIdentificationType	ProductIdentificationType	Enum	1	Type of product identification. Name · Value · Description ISIN · 1 · Financial instrument's ISIN.
mic	MicCode	Array (AnsiChar)	4	Market structure's Market Identifier Code (MIC) as specified in ISO 10383.
currency	Currency	Enum	2	Currency (e.g. USD).
instrumentPacket	u64	Primitive	8	Instrument packet in index portfolio / of information product. Number of units (usually stocks/shares) of a given instrument in the index portfolio or in the information product.

11.2.14. INDEXSUMMARY:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument.
openingIndexValue	IndexValue	Alias (Price)	8	Opening index value

Name	Data type	Kind	Length	Description
sessionLow	IndexValue	Alias (Price)	8	Lowest value of the day.
sessionHigh	IndexValue	Alias (Price)	8	Highest value of the day.
closingIndexValue	IndexValue	Alias (Price)	8	Closing index value.
sessionAvg	IndexValue	Alias (Price)	8	Average value of the day.
pctChangeIndexValPrevSession	PercentageChange	Alias (Price)	8	Percentage change of the index value in relation to the value of the closing index from the previous session.
pctChangeIndexValPrevYear	PercentageChange	Alias (Price)	8	Percentage change of the index value in relation to its value at the end of the previous year
closingCapitalisationPortfolio	Value	Alias (i64)	8	Capitalisation of the portfolio for the closing index expressed in the currency of the index, on basis of which the closing index is calculated.
indIndexOpeningPortfolio	PercentageChange	Alias (Price)	8	Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.

11.2.15. INDEXATIONTABLEENTRY:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
productId	ElementId	Alias (u32)	4	Product ID.
indexationTableEntryDate	Date	Alias (u32)	4	Indexation table entry date.
indexationTableEntryValue	Number	Alias (i64)	8	Indexation table entry value.

11.2.16. INSTRUMENT:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>productType</td><td>ProductType</td><td>Enum</td><td>1</td><td>Type of the product.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>FinancialProductShare</td><td>1</td><td>Equity</td></tr>
<tr><td>FinancialProductBond</td><td>2</td><td>Fixed income</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table>
<tr><td>FinancialProductDerivativeFutures</td><td>3</td><td>Futures</td></tr>
<tr><td>FinancialProductDerivativeOptions</td><td>4</td><td>Options</td></tr>
<tr><td>FinancialProductIndex</td><td>5</td><td>Index</td></tr>
<tr><td>FinancialProductCurrency</td><td>6</td><td>Currency</td></tr>
</table></td></tr>
<tr><td>firstTradingDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>First trading day of the financial instrument.</td></tr>
<tr><td>lastTradingDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Last trading day of the financial instrument.</td></tr>
<tr><td>productId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Reference to financial product definition.</td></tr>
<tr><td>currency</td><td>Currency</td><td>Enum</td><td>2</td><td>Trading currency (e.g. USD).</td></tr>
<tr><td>lotSize</td><td>LotSize</td><td>Alias (u32)</td><td>4</td><td>Lot size for the instrument.</td></tr>
<tr><td>priceExpressionType</td><td>PriceExpressionType</td><td>Enum</td><td>1</td><td>Price expression type for the financial instrument.<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Price</td><td>1</td><td>Price expressed as absolute value.</td></tr>
<tr><td>Percentage</td><td>2</td><td>Price expressed as percentage.</td></tr>
</table></td></tr>
<tr><td>calendarId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of trading calendar for the financial instrument.</td></tr>
<tr><td>marketStructureId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the financial instrument's market segment.</td></tr>
<tr><td>tickTableId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the tick table used for the financial instrument.</td></tr>
<tr><td>collarGroupId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Collar group ID.</td></tr>
<tr><td>tradingScheduleId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Trading schedule ID.</td></tr>
<tr><td>code</td><td>InstrumentCode</td><td>Array (AnsiChar)</td><td>64</td><td>Financial instrument code.</td></tr>
<tr><td>mic</td><td>MicCode</td><td>Array (AnsiChar)</td><td>4</td><td>Market structure's Market Identifier Code (MIC) as specified in ISO 10383.</td></tr>
<tr><td>nominalValue</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Nominal value of the financial instrument.</td></tr>
<tr><td>nominalValueType</td><td>NominalValueType</td><td>Enum</td><td>1</td><td>Type of the product's nominal value (no nominal, constant or unknown).<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>Constant</td><td>2</td><td>Indicates that the security has a constant nominal value.</td></tr>
</table></td></tr>
<tr><td>multiplier</td><td>Multiplier</td><td>Alias (Number)</td><td>8</td><td>Product value multiplier.</td></tr>
<tr><td>strikePrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Product strike price.</td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>productIdentification</td><td>ProductIdentification</td><td>Array (AnsiChar)</td><td>30</td><td>Product identification, e.g. ISIN number.</td></tr>
<tr><td>productIdentificationType</td><td>ProductIdentificationType</td><td>Enum</td><td>1</td><td>Type of product identification.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>ISIN</td><td>1</td><td>Financial instrument's ISIN.</td></tr></table></td></tr>
<tr><td>settlementCalendarId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Settlement calendar ID.</td></tr>
<tr><td>bondCouponType</td><td>CouponType</td><td>Enum</td><td>1</td><td>Coupon type for a bond.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NotApplicable</td><td>1</td><td>NotApplicable.</td></tr><tr><td>Zero</td><td>2</td><td>Zero coupon.</td></tr><tr><td>Fixed</td><td>3</td><td>Fixed coupon.</td></tr><tr><td>Floating</td><td>4</td><td>Floating coupon.</td></tr><tr><td>Indexed</td><td>5</td><td>Indexed coupon.</td></tr></table></td></tr>
<tr><td>preTradeCheckMinPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Minimum order price for pre-trade check.</td></tr>
<tr><td>preTradeCheckMaxPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Maximum order price for pre-trade check.</td></tr>
<tr><td>preTradeCheckMinQuantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Minimum volume for pre-trade check.</td></tr>
<tr><td>preTradeCheckMaxQuantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Maximum volume for pre-trade check.</td></tr>
<tr><td>preTradeCheckMaxValue</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Maximum value for pre-trade check.</td></tr>
<tr><td>preTradeCheckMinValue</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Minimum value for pre-trade check.</td></tr>
<tr><td>liquidity</td><td>bool</td><td>Primitive</td><td>1</td><td>Liquidity flag, true - liquid, false - Illiquid.</td></tr>
<tr><td>marketModelType</td><td>MarketModelType</td><td>Enum</td><td>1</td><td>Market Model.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NotApplicable</td><td>1</td><td>Not applicable</td></tr><tr><td>CLOB</td><td>2</td><td>A Central Limit Order Book (CLOB)</td></tr><tr><td>BLOCK</td><td>3</td><td>BLOCK</td></tr><tr><td>HYBRID</td><td>4</td><td>HYBRID market model</td></tr><tr><td>CROSS</td><td>5</td><td>CROSS</td></tr><tr><td>IPO</td><td>6</td><td>Initial public offering</td></tr></table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>issuer</td><td>Issuer</td><td>Array (AnsiChar)</td><td>150</td><td>Name of the issuer.</td></tr>
<tr><td>issuerRegCountry</td><td>Country</td><td>Enum</td><td>2</td><td>Identifier of the country of the registration.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>AUS</td><td>36</td><td>Australia</td></tr>
<tr><td>AUT</td><td>40</td><td>Austria</td></tr>
<tr><td>BEL</td><td>56</td><td>Belgium</td></tr>
<tr><td>BGR</td><td>100</td><td>Bulgaria</td></tr>
<tr><td>CAN</td><td>124</td><td>Canada</td></tr>
<tr><td>CYP</td><td>196</td><td>Cyprus</td></tr>
<tr><td>CZE</td><td>203</td><td>Czechia</td></tr>
<tr><td>EST</td><td>233</td><td>Estonia</td></tr>
<tr><td>FRA</td><td>250</td><td>France</td></tr>
<tr><td>DEU</td><td>276</td><td>Germany</td></tr>
<tr><td>HUN</td><td>348</td><td>Hungary</td></tr>
<tr><td>IRL</td><td>372</td><td>Ireland</td></tr>
<tr><td>ISR</td><td>376</td><td>Israel</td></tr>
<tr><td>ITA</td><td>380</td><td>Italy</td></tr>
<tr><td>LTU</td><td>440</td><td>Lithuania</td></tr>
<tr><td>LUX</td><td>442</td><td>Luxembourg</td></tr>
<tr><td>NLD</td><td>528</td><td>Netherlands (the)</td></tr>
<tr><td>POL</td><td>616</td><td>Poland</td></tr>
<tr><td>PRT</td><td>620</td><td>Portugal</td></tr>
<tr><td>ROU</td><td>642</td><td>Romania</td></tr>
<tr><td>SVK</td><td>703</td><td>Slovakia</td></tr>
<tr><td>SVN</td><td>705</td><td>Slovenia</td></tr>
<tr><td>ESP</td><td>724</td><td>Spain</td></tr>
<tr><td>SWE</td><td>752</td><td>Sweden</td></tr>
<tr><td>UKR</td><td>804</td><td>Ukraine</td></tr>
<tr><td>GBR</td><td>826</td><td>United Kingdom of Great Britain and Northern Ireland (the)</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table><tr><td>USA</td><td>840</td><td>United States of America (the)</td></tr></table></td></tr>
<tr><td>underlyingInstrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Identifier of the underlying instrument.</td></tr>
<tr><td>productCode</td><td>Code</td><td>Array (AnsiChar)</td><td>16</td><td>Unique code of the product.</td></tr>
<tr><td>cfi</td><td>CfiCode</td><td>Array (AnsiChar)</td><td>6</td><td>CFI (Classification of Financial Instrument) code.</td></tr>
<tr><td>fisn</td><td>FisnCode</td><td>Array (AnsiChar)</td><td>35</td><td>Product FISN (Financial Instrument Short Name) code.</td></tr>
<tr><td>issueSize</td><td>u64</td><td>Primitive</td><td>8</td><td>Size of the issue (depends on the issue size type)</td></tr>
<tr><td>nominalCurrency</td><td>Currency</td><td>Enum</td><td>2</td><td>Identifier of the nominal currency (ISO-4217).</td></tr>
<tr><td>usIndicator</td><td>UsIndicator</td><td>Enum</td><td>1</td><td>Identifier of the us indicator.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NotApplicable</td><td>1</td><td>No trading restrictions.</td></tr><tr><td>RegulationS</td><td>2</td><td>Trading restricted due to US Regulation S.</td></tr><tr><td>RegulationSPlusRule144A</td><td>3</td><td>Trading restricted due to US Regulation S with Rule 144A.</td></tr></table></td></tr>
<tr><td>expiryDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Expiration date.</td></tr>
<tr><td>versionNumber</td><td>u8</td><td>Primitive</td><td>1</td><td>Version of the derivative. Version is incremented when corporate event adjusting underlying closing price occurs.</td></tr>
<tr><td>settlementType</td><td>SettlementType</td><td>Enum</td><td>1</td><td>Identifier of settlement type.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NotApplicable</td><td>1</td><td>Not applicable.</td></tr><tr><td>CashSettlement</td><td>2</td><td>Derivatives are settled in cash.</td></tr><tr><td>PhysicalDelivery</td><td>3</td><td>Derivatives are settled as physical delivery of the underlying.</td></tr></table></td></tr>
<tr><td>optionType</td><td>OptionType</td><td>Enum</td><td>1</td><td>Identifier of option type.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NotApplicable</td><td>1</td><td>Not applicable.</td></tr><tr><td>Call</td><td>2</td><td>Call option.</td></tr><tr><td>Put</td><td>3</td><td>Put option.</td></tr></table></td></tr>
<tr><td>excerciseType</td><td>ExcerciseType</td><td>Enum</td><td>1</td><td>Identifier of exercise style.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>AMER</td><td>1</td><td>AMER</td></tr></table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table><tr><td>EURO</td><td>2</td><td>EURO</td></tr><tr><td>NA</td><td>3</td><td>Not applicable</td></tr></table></td></tr>
<tr><td>productName</td><td>Name</td><td>Array (AnsiChar)</td><td>50</td><td>Name of the product.</td></tr>
<tr><td>referencePrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Reference price</td></tr>
<tr><td>status</td><td>InstrumentStatus</td><td>Enum</td><td>1</td><td>Instrument status.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Active</td><td>1</td><td>Financial instrument is active.</td></tr><tr><td>Inactive</td><td>2</td><td>Financial instrument is inactive and is not trading.</td></tr><tr><td>MarketOperationsSuspension</td><td>3</td><td>Manual suspension by Market Operations.</td></tr><tr><td>OutsideCollarsStatic</td><td>4</td><td>Outside static trade price collars.</td></tr><tr><td>OutsideCollarsDynamic</td><td>5</td><td>Outside dynamic trade price collars.</td></tr><tr><td>RegulatorySuspension</td><td>6</td><td>Regulatory suspension</td></tr><tr><td>TechnicalHalt</td><td>7</td><td>TechnicalHalt.</td></tr><tr><td>HybridNoQuotes</td><td>8</td><td>Hybrid no valid quotes.</td></tr><tr><td>HybridPause</td><td>9</td><td>Hybrid pause.</td></tr></table></td></tr>
<tr><td>initialPhaseId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Id of starting phase for the instrument</td></tr>
</table>

11.2.17. INSTRUMENTSTATUSCHANGE:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>tradingPhaseId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Trading phase ID.</td></tr>
<tr><td>status</td><td>InstrumentStatus</td><td>Enum</td><td>1</td><td>Financial instrument status.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Active</td><td>1</td><td>Financial instrument is active.</td></tr><tr><td>Inactive</td><td>2</td><td>Financial instrument is inactive and is not trading.</td></tr><tr><td>MarketOperationsSuspension</td><td>3</td><td>Manual suspension by Market Operations.</td></tr></table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table>
<tr><td>OutsideCollarsStatic</td><td>4</td><td>Outside static trade price collars.</td></tr>
<tr><td>OutsideCollarsDynamic</td><td>5</td><td>Outside dynamic trade price collars.</td></tr>
<tr><td>RegulatorySuspension</td><td>6</td><td>Regulatory suspension</td></tr>
<tr><td>TechnicalHalt</td><td>7</td><td>TechnicalHalt.</td></tr>
<tr><td>HybridNoQuotes</td><td>8</td><td>Hybrid no valid quotes.</td></tr>
<tr><td>HybridPause</td><td>9</td><td>Hybrid pause.</td></tr>
</table></td></tr>
<tr><td>stressedMarket</td><td>bool</td><td>Primitive</td><td>1</td><td>Stressed market conditions is called for when instrument experience high and short term intraday volatility.</td></tr>
</table>

11.2.18. INSTRUMENTSUMMARY:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Identifier of the CLOB instrument.</td></tr>
<tr><td>lastTradedPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Last Traded Price (LTP).</td></tr>
<tr><td>closingPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Closing Price (CP).</td></tr>
<tr><td>closingPriceType</td><td>ClosingPriceType</td><td>Enum</td><td>1</td><td>Closing Price Type.<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>LTP</td><td>1</td><td>0 = LTP</td></tr>
<tr><td>LastACP</td><td>2</td><td>1 = Last ACP</td></tr>
<tr><td>FairValue</td><td>3</td><td>2 = Fair Value</td></tr>
<tr><td>DailySettlementPrice</td><td>4</td><td>3 = Daily Settlement Price</td></tr>
<tr><td>FinalSettlementPrice</td><td>5</td><td>4 = Final Settlement Price</td></tr>
</table></td></tr>
<tr><td>adjustedClosingPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Adjusted Closing Price (ACP).</td></tr>
<tr><td>adjustedClosingPriceReason</td><td>AdjustedClosingPriceReason</td><td>Enum</td><td>1</td><td>Adjusted Closing Price Reason.<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Regular</td><td>1</td><td>0 = Regular</td></tr>
<tr><td>Dividend</td><td>2</td><td>1 = Dividend</td></tr>
<tr><td>IssueRight</td><td>3</td><td>2 = Issue Right</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table><tr><td>Split</td><td>4</td><td>3 = Split</td></tr><tr><td>ReverseSplit</td><td>5</td><td>4 = Reverse Split</td></tr><tr><td>Bonus</td><td>6</td><td>5 = Bonus</td></tr><tr><td>SpinOff</td><td>7</td><td>6 = Spin-Off</td></tr></table></td></tr>
<tr><td>pctChange</td><td>PercentageChange</td><td>Alias (Price)</td><td>8</td><td>Percentage change.</td></tr>
<tr><td>VWAP</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Volume-weighted average price.</td></tr>
<tr><td>noTrades</td><td>u64</td><td>Primitive</td><td>8</td><td>Total number of transations on the current trading day.</td></tr>
<tr><td>totalVolume</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Total transaction volume.</td></tr>
<tr><td>totalValue</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Total transaction value.</td></tr>
<tr><td>openingPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>The price of the first trade on the current trading day.</td></tr>
<tr><td>maxPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Highest price of the instrument on the current trading day</td></tr>
<tr><td>minPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Lowest price of the instrument on the current trading day</td></tr>
</table>

11.2.19. LOGIN:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
connectionId	ConnectionId	Alias (u16)	2	ID of the connection.
token	Token	Array (AnsiChar)	8	Client token.

11.2.20. LOGINRESPONSE:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>result</td><td>LoginResult</td><td>Enum</td><td>1</td><td>see the LoginResult description.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Ok</td><td>1</td><td>Successful login.</td></tr></table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table><tr><td>InvalidToken</td><td>2</td><td>Authorization failure.</td></tr><tr><td>AlreadyLoggedIn</td><td>3</td><td>Already logged in.</td></tr></table></td></tr>
</table>

11.2.21. LOGOUT:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.

11.2.22. MARKETSTRUCTURE:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>id</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the market structure.</td></tr>
<tr><td>parentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the market structure's parent market structure if the market structure is a sub-structure. Set to '0' (zero) if no parent market structure.</td></tr>
<tr><td>mic</td><td>MicCode</td><td>Array (AnsiChar)</td><td>4</td><td>Market structure's Market Identifier Code (MIC) as specified in ISO 10383.</td></tr>
<tr><td>name</td><td>Name</td><td>Array (AnsiChar)</td><td>50</td><td>Market structure name assigned by market operator.</td></tr>
<tr><td>marketModelType</td><td>MarketModelType</td><td>Enum</td><td>1</td><td>Market model type.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>NotApplicable</td><td>1</td><td>Not applicable</td></tr><tr><td>CLOB</td><td>2</td><td>A Central Limit Order Book (CLOB)</td></tr><tr><td>BLOCK</td><td>3</td><td>BLOCK</td></tr><tr><td>HYBRID</td><td>4</td><td>HYBRID market model</td></tr><tr><td>CROSS</td><td>5</td><td>CROSS</td></tr><tr><td>IPO</td><td>6</td><td>Initial public offering</td></tr></table></td></tr>
</table>

11.2.23. NEWS:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header
marketStructureId	ElementId	Alias (u32)	4	ID of the instrument's market structure.

title	AnsiChar	Alias (u8)	1	News title, unique per message.
entryNumber	u8	Primitive	1	News entry number.
total	u8	Primitive	1	Total number of news entries.
text	AnsiChar	Alias (u8)	1	News text.

11.2.24. ORDERADD:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>publicOrderId</td><td>PublicOrderId</td><td>Alias (u64)</td><td>8</td><td>Order identifier (ID).</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>side</td><td>OrderSide</td><td>Enum</td><td>1</td><td>Order side (buy or sell).<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Buy</td><td>1</td><td>Buy order.</td></tr><tr><td>Sell</td><td>2</td><td>Sell order.</td></tr></table></td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Order (limit) price.</td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Order quantity.</td></tr>
</table>

11.2.25. ORDERBOOKEVENT:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the instrument.</td></tr>
<tr><td>eventType</td><td>OrderBookEventType</td><td>Enum</td><td>1</td><td>Type/Kind of book event<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Clear</td><td>1</td><td>Clear order book</td></tr><tr><td>ResendStart</td><td>2</td><td>Order book resend start (resend will begin shortly)</td></tr><tr><td>ResendEnd</td><td>3</td><td>Order book resend end (resend is done)</td></tr></table></td></tr>
</table>

11.2.26. ORDERCOLLARS:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>collarType</td><td>CollarType</td><td>Enum</td><td>1</td><td>Collars type.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Static</td><td>1</td><td></td></tr>
<tr><td>Dynamic</td><td>2</td><td></td></tr>
</table></td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>The reference price.</td></tr>
<tr><td>lowerBid</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Value of the lower bid Price collar.</td></tr>
<tr><td>upperBid</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Value of the upper bid Price collar.</td></tr>
<tr><td>lowerAsk</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Value of the lower ask Price collar.</td></tr>
<tr><td>upperAsk</td><td>Bound</td><td>Alias (Number)</td><td>8</td><td>Value of the upper ask Price collar.</td></tr>
</table>

11.2.27. ORDERDELETE:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
publicOrderId	PublicOrderId	Alias (u64)	8	Order identifier (ID).

11.2.28. ORDEREXECUTE:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
quantity	Quantity	Alias (u64)	8	Remaining un-executed order quantity.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
publicOrderId	PublicOrderId	Alias (u64)	8	Order identifier (ID).
executionId	TradeId	Alias (u32)	4	ID of the underlying trade (equal to TradeID in Trade.)
executionPrice	Price	Alias (Number)	8	Price at which the order was executed.
executionQuantity	Quantity	Alias (u64)	8	The order's executed quantity.

11.2.29. ORDERMODIFY:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>publicOrderId</td><td>PublicOrderId</td><td>Alias (u64)</td><td>8</td><td>Order identifier (ID).</td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>New order price.</td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>New order quantity.</td></tr>
<tr><td>priorityFlag</td><td>PriorityFlag</td><td>Enum</td><td>1</td><td>Priority loss flag.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Lost</td><td>1</td><td>Order modification caused loss of priority.</td></tr>
<tr><td>Retained</td><td>2</td><td>Order modification did not cause loss of priority.</td></tr>
</table>
</td></tr>
</table>

11.2.30. PRICELEVELSNAPSHOT:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>maxDepth</td><td>u8</td><td>Primitive</td><td>1</td><td>The number of BBO levels contained in the message.</td></tr>
<tr><td>buy</td><td>PriceLevels</td><td>Array (PriceLevel)</td><td>310</td><td>Price levels for buy side.
PriceLevel:
<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>isEncrypted</td><td>bool</td><td>Primitive</td><td>1</td><td>A boolean value specifying whether the level is encrypted.</td></tr>
<tr><td>encryptionOffset</td><td>u64</td><td>Primitive</td><td>8</td><td>Encryption byte offset.</td></tr>
<tr><td>encryptionKeyId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Encryption Key ID - reference to a EncryptionKey message.</td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Price level in currency units.</td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Quantity of securities available at a given price level.</td></tr>
<tr><td>orderCount</td><td>OrderCount</td><td>Alias (u16)</td><td>2</td><td>Number of open orders at a given price level.</td></tr>
</table>
</td></tr>
<tr><td>sell</td><td>PriceLevels</td><td>Array (PriceLevel)</td><td>310</td><td>Price levels for sell side.
PriceLevel:</td></tr>
</table>

<table>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>isEncrypted</td><td>bool</td><td>Primitive</td><td>1</td><td>A boolean value specifying whether the level is encrypted.</td></tr>
<tr><td>encryptionOffset</td><td>u64</td><td>Primitive</td><td>8</td><td>Encryption byte offset.</td></tr>
<tr><td>encryptionKeyId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Encryption Key ID - reference to a EncryptionKey message.</td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Price level in currency units.</td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Quantity of securities available at a given price level.</td></tr>
<tr><td>orderCount</td><td>OrderCount</td><td>Alias (u16)</td><td>2</td><td>Number of open orders at a given price level.</td></tr>
</table>
</td></tr>
</table>

11.2.31. PRICEUPDATE:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Instrument to which the price update refers.</td></tr>
<tr><td>priceType</td><td>PriceType</td><td>Enum</td><td>1</td><td>Price type.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>ReferencePrice</td><td>1</td><td></td></tr>
<tr><td>MidPoint</td><td>2</td><td></td></tr>
</table>
</td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Indicates the updated price.</td></tr>
</table>

11.2.32. REALTIMEINDEX:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the instrument.</td></tr>
<tr><td>indexLevelCode</td><td>IndexLevelCode</td><td>Enum</td><td>1</td><td>Level of the index.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>PreviousClosingValue</td><td>1</td><td>Closing index value from previous session (reference index).</td></tr>
<tr><td>TheoreticalValueBeforeOpening</td><td>2</td><td>Theoretical opening index before opening the session.</td></tr>
<tr><td>ValueBeforeOpening</td><td>3</td><td>Theoretical value of the index before its opening.</td></tr>
<tr><td>OpeningValue</td><td>4</td><td>Index opening value.</td></tr>
<tr><td>CurrentValue</td><td>5</td><td>Current index value.</td></tr>
</table>
</td></tr>
</table>

Name	Data type	Kind	Length	Description
				PreClosingValue 6 Preliminary closing value. ClosingValue 7 Closing index value. ValueWithoutOpening 8 Index value without opening.
numOfActiveInstruments	u16	Primitive	2	Number of instruments – participants of the index, for which the quotation was set during the session.
indIndexOpeningPortfolio	PercentageChange	Alias (Price)	8	Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.
indexValue	IndexValue	Alias (Price)	8	The value of the last level for the index according to the definition provided in the indexLevelCode.
pctChangeIndexValPrevSession	PercentageChange	Alias (Price)	8	Percentage change of the index value in relation to the value of the closing index from the previous session.
tradingValue	Value	Alias (i64)	8	Value of trading in the instruments from the given index.
sessionLow	IndexValue	Alias (Price)	8	Lowest value of the day.
sessionHigh	IndexValue	Alias (Price)	8	Highest value of the day.
indexValueBid	IndexValue	Alias (Price)	8	Value of the buy index based on the best sell offers of instruments.
indexValueAsk	IndexValue	Alias (Price)	8	Value of the sell index based on the best sell offers of instruments.
midSpreadIndex	IndexValue	Alias (Price)	8	Mean of indexValueBid and indexValueAsk.
differenceCentralSpread	IndexValue	Alias (Price)	8	The point difference between midSpreadIndex and indexValue.

11.2.33. REFERENCEDATAEND:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.

11.2.34. REFERENCEDATASTART:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.

11.2.35. SESSIONSUMMARY:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Identifier of the CLOB instrument.</td></tr>
<tr><td>lastTradedPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Last Traded Price (LTP).</td></tr>
<tr><td>closingPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Closing Price (CP).</td></tr>
<tr><td>closingPriceType</td><td>ClosingPriceType</td><td>Enum</td><td>1</td><td>Closing Price Type.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>LTP</td><td>1</td><td>0 = LTP</td></tr>
<tr><td>LastACP</td><td>2</td><td>1 = Last ACP</td></tr>
<tr><td>FairValue</td><td>3</td><td>2 = Fair Value</td></tr>
<tr><td>DailySettlementPrice</td><td>4</td><td>3 = Daily Settlement Price</td></tr>
<tr><td>FinalSettlementPrice</td><td>5</td><td>4 = Final Settlement Price</td></tr>
</table></td></tr>
<tr><td>pctChange</td><td>PercentageChange</td><td>Alias (Price)</td><td>8</td><td>Percentage change.</td></tr>
<tr><td>vwap</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Volume-weighted average price.</td></tr>
<tr><td>noTrades</td><td>u64</td><td>Primitive</td><td>8</td><td>Total number of transations on the current trading day.</td></tr>
<tr><td>totalVolume</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Total transaction volume.</td></tr>
<tr><td>totalValue</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Total transaction value.</td></tr>
<tr><td>openingPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>The price of the first trade on the current trading day.</td></tr>
<tr><td>maxPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Highest price of the instrument on the current trading day</td></tr>
<tr><td>minPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Lowest price of the instrument on the current trading day</td></tr>
<tr><td>productIdentificationType</td><td>ProductIdentificationType</td><td>Enum</td><td>1</td><td>Type of product identification.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>ISIN</td><td>1</td><td>Financial instrument's ISIN.</td></tr>
</table></td></tr>
<tr><td>productIdentification</td><td>ProductIdentification</td><td>Array (AnsiChar)</td><td>30</td><td>Product identification, e.g. ISIN number.</td></tr>
</table>

Name	Data type	Kind	Length	Description
AccumulatedInterest	Number	Alias (i64)	8	Accumulated interest on the bonds or for mortgage-backed bonds on the day of settling the transaction.
interestRate	Number	Alias (i64)	8	Interest rate is determined on the basis of WIBOR/WIBID/WIRON rates for each expiration date of options and futures.
referencePrice	Price	Alias (Number)	8	Initial reference price (at the start of session day).
settlementPrice	Price	Alias (Number)	8	Settlement price.
settlementValue	Price	Alias (Number)	8	Settlement value.
currency	Currency	Enum	2	Price currency (e.g. USD).
openPositions	u32	Primitive	4	Number of open positions after the end of the session.
sessionDate	Date	Alias (u32)	4	Stock Exchange session date.
endTradingDate	Date	Alias (u32)	4	End of trading date.
lastTradeDate	Date	Alias (u32)	4	The date of execution of the last trade for the given instrument.
numberOfInstruments	u64	Primitive	8	Number of instruments admitted to trading.
tradingValueCurrency	Value	Alias (i64)	8	The field comprises the trading value expressed in trading currency.
blockNoTrades	u64	Primitive	8	The total number of block trades concluded on a particular instrument during the current trading session.
blockVolume	Quantity	Alias (u64)	8	The total turnover volume of block trades concluded on a particular instrument during the current trading session.
blockValue	Value	Alias (i64)	8	The total turnover value of block trades concluded on a particular instrument during the current trading session.
blockMinPrice	Price	Alias (Number)	8	Minimum price of block trades in the instrument during the current trading session. If no block trades were concluded during the trading session, the field is completed with zeros.
blockMaxPrice	Price	Alias (Number)	8	Maximum price of block trades in the instrument during the current trading session. If no block trades were concluded during the trading session, the field is completed with zeros.
vwapBlockTrade	Price	Alias (Number)	8	Volume weighted average price of block trades for a particular instrument during a particular trading session.
status	InstrumentStatus	Enum	1	Financial instrument status. <table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Active</td><td>1</td><td>Financial instrument is active.</td></tr><tr><td>Inactive</td><td>2</td><td>Financial instrument is inactive and is not trading.</td></tr><tr><td>MarketOperationsSuspension</td><td>3</td><td>Manual suspension by Market Operations.</td></tr><tr><td>OutsideCollarsStatic</td><td>4</td><td>Outside static trade price collars.</td></tr></table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>OutsideCollarsDynamic</td><td>5</td><td>Outside dynamic trade price collars.</td></tr>
<tr><td>RegulatorySuspension</td><td>6</td><td>Regulatory suspension</td></tr>
<tr><td>TechnicalHalt</td><td>7</td><td>TechnicalHalt.</td></tr>
<tr><td>HybridNoQuotes</td><td>8</td><td>Hybrid no valid quotes.</td></tr>
<tr><td>HybridPause</td><td>9</td><td>Hybrid pause.</td></tr>
</table>
</td></tr>
<tr><td>sector</td><td>u16</td><td>Primitive</td><td>2</td><td>Defines the sector of the economy that the company belongs to. Possible values for shares, the field assumes were described in the WATS Market Data documetation.</td></tr>
<tr><td>market</td><td>Market</td><td>Enum</td><td>1</td><td>Defines the market the instrument belongs to.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Primary</td><td>1</td><td>Primary market.</td></tr>
<tr><td>Parallel</td><td>2</td><td>Parallel market.</td></tr>
<tr><td>NewConnectPriceDriven</td><td>3</td><td>New Connect Market (price driven market).</td></tr>
<tr><td>NewConnectOrderDriven</td><td>4</td><td>New Connect Market (order driven market).</td></tr>
<tr><td>CatalystRegulated</td><td>5</td><td>Catalyst Regulated Market.</td></tr>
<tr><td>CatalystAso</td><td>6</td><td>Catalyst ASO.</td></tr>
<tr><td>Other</td><td>7</td><td>Other market.</td></tr>
</table>
</td></tr>
<tr><td>markerPriceChange</td><td>ChangeIndicator</td><td>Enum</td><td>1</td><td>The field contains the market of the percentage change of the instrument closing price from the current session in relation to the reference price.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Increase</td><td>1</td><td>Increase.</td></tr>
<tr><td>Decrease</td><td>2</td><td>Decrease.</td></tr>
<tr><td>NoChange</td><td>3</td><td>No change.</td></tr>
<tr><td>NotApplicable</td><td>4</td><td>Not Applicable.</td></tr>
</table>
</td></tr>
<tr><td>marketStructureId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the financial instrument's market segment.</td></tr>
<tr><td>lowerLiquidity</td><td>bool</td><td>Primitive</td><td>1</td><td>The positive field value (true) informs if the company was qualified to Lower Liquidity Space. A negative value has the opposite meaning.</td></tr>
<tr><td>multiplier</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Instrument multiplier.</td></tr>
<tr><td>impliedVolatility</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Implied volatility.</td></tr>
<tr><td>dividendRate</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Dividend rate for the company is given on the basis of dividends paid by companies being the base instrument for futures and options.</td></tr>
<tr><td>delta</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Value of Delta indicator from the current session.</td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>gamma</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Value of Gamma indicator from the current session.</td></tr>
<tr><td>rho</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Value of Rho indicator from the current session.</td></tr>
<tr><td>theta</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Value of Theta indicator from the current session.</td></tr>
<tr><td>vega</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Value of Vega indicator from the current session.</td></tr>
<tr><td>volatility</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Volatility to be calculated is based on the implied volatility (provided in field ImpliedVolatility) according to algorithm worked out by the WSE, which is available on the website: www.opcje.gpw.pl.</td></tr>
<tr><td>optionsStrikePrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Option product strike price.</td></tr>
<tr><td>dividend</td><td>bool</td><td>Primitive</td><td>1</td><td>The positive field value (true) informs that the instruments are traded the first time after determining the right to dividend. A negative value has the opposite meaning.</td></tr>
<tr><td>subscriptionRight</td><td>bool</td><td>Primitive</td><td>1</td><td>The positive field value (true) informs that the instruments are traded the first time after determining the subscription right. A negative value has the opposite meaning.</td></tr>
<tr><td>interimDividendRight</td><td>bool</td><td>Primitive</td><td>1</td><td>The positive field value (true) informs that the instruments are traded the first time after determining the interim dividend right. A negative value has the opposite meaning.</td></tr>
<tr><td>split</td><td>bool</td><td>Primitive</td><td>1</td><td>The positive field value (true) informs that the instruments are traded the first time after the change of the nominal value of shares. A negative value has the opposite meaning.</td></tr>
<tr><td>quotationSystem</td><td>QuotationSystem</td><td>Enum</td><td>1</td><td>The field informing about the system of listing the instrument.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>SinglePriceSingleFixing</td><td>1</td><td>Single price quotation system with a single fixing.</td></tr>
<tr><td>ContinuousTrading</td><td>2</td><td>Continuous trading.</td></tr>
<tr><td>SinglePriceTwoFixings</td><td>3</td><td>Single price quotation system with two fixings.</td></tr>
</table></td></tr>
<tr><td>liquiditySupportPge</td><td>bool</td><td>Primitive</td><td>1</td><td>The field informs if the company entered to Liquidity Support Programme. True - participation in the liquidity Programme, false - no participation in the liquidity Programme.</td></tr>
</table>

11.2.36. STARTOFTECHNICALSESSION:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
sessionId	SessionId	Alias (u16)	2	ID of the session.

11.2.37. TICKTABLEENTRY:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.

Name	Data type	Kind	Length	Description
tickSize	TickSize	Alias (Number)	8	Size of the tick.
lowerBound	Bound	Alias (Number)	8	Tick size lower bound.
tickTableId	ElementId	Alias (u32)	4	Tick table ID.

11.2.38. TOPPRICELEVELUPDATE:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>BBO level 1 price level in currency units.</td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>BBO level 1 quantity at price level.</td></tr>
<tr><td>side</td><td>OrderSide</td><td>Enum</td><td>1</td><td>Side (buy or sell).<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Buy</td><td>1</td><td>Buy order.</td></tr><tr><td>Sell</td><td>2</td><td>Sell order.</td></tr></table></td></tr>
<tr><td>orderCount</td><td>OrderCount</td><td>Alias (u16)</td><td>2</td><td>Number of open orders at BBO level 1.</td></tr>
</table>

11.2.39. TRADE:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
settlementValue	Value	Alias (i64)	8	Settlement value.
settlementDate	Date	Alias (u32)	4	Settlement date.
tradingTimestamp	Timestamp	Alias (u64)	8	Date and time when the transaction was executed.
publicProductIdentification.productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification, for example its ISIN code.

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>publicProductIdentification.productIdentificationType</td><td>ProductIdentificationType</td><td>Enum</td><td>1</td><td>Type of product identification.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>ISIN</td><td>1</td><td>Financial instrument's ISIN.</td></tr></table></td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Trade price.</td></tr>
<tr><td>priceCurrency</td><td>Currency</td><td>Enum</td><td>2</td><td>Currency in which the price is expressed (applicable if the price is expressed as monetary value).</td></tr>
<tr><td>priceNotation</td><td>PriceExpressionType</td><td>Enum</td><td>1</td><td>Indication as to whether the price is expressed in monetary value, in percentage or in yield.<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>Price</td><td>1</td><td>Price expressed as absolute value.</td></tr><tr><td>Percentage</td><td>2</td><td>Price expressed as percentage.</td></tr></table></td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Trade quantity.</td></tr>
<tr><td>nominalValue</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Nominal value.</td></tr>
<tr><td>nominalCurrency</td><td>Currency</td><td>Enum</td><td>2</td><td>Currency in which the notional is denominated.</td></tr>
<tr><td>mic</td><td>MicCode</td><td>Array (AnsiChar)</td><td>4</td><td>Identification of the venue where the transaction was executed.</td></tr>
<tr><td>publicationTimestamp</td><td>Timestamp</td><td>Alias (u64)</td><td>8</td><td>Date and time when the transaction was published by a trading venue.</td></tr>
<tr><td>tradeId</td><td>TradeId</td><td>Alias (u32)</td><td>4</td><td>ID of the trade.</td></tr>
<tr><td>tradeToBeCleared</td><td>bool</td><td>Primitive</td><td>1</td><td>Code to identify whether the transaction will be cleared.</td></tr>
<tr><td>mmtMarketMechanism</td><td>MmtMarketMechanism</td><td>Enum</td><td>1</td><td>MMT Market Mechanism<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>CentralLimitOrderBook</td><td>1</td><td>1 = Central Limit Order Book</td></tr><tr><td>QuoteDrivenMarket</td><td>2</td><td>2 = Quote Driven Market</td></tr><tr><td>DarkOrderBook</td><td>3</td><td>3 = Dark Order Book</td></tr><tr><td>OffBook</td><td>4</td><td>4 = Off Book</td></tr></table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Lengt h</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table>
<tr><td>PeriodicAuction</td><td>5</td><td>5 = Periodic Auction</td></tr>
<tr><td>RequestForQuotes</td><td>6</td><td>6 = Request For Quotes</td></tr>
<tr><td>Other</td><td>7</td><td>7 = Any other including Hybrid</td></tr>
</table></td></tr>
<tr><td>mmtTradingMode</td><td>MmtTradingMode</td><td>Enum</td><td>1</td><td>MMT Trading Mode<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>UndefinedAuction</td><td>1</td><td>1 = Undefined Auction</td></tr>
<tr><td>ScheduledOpeningAuction</td><td>2</td><td>O = Scheduled Opening Auction</td></tr>
<tr><td>ScheduledClosingAuction</td><td>3</td><td>K = Scheduled Closing Auction</td></tr>
<tr><td>ScheduledIntradayAuction</td><td>4</td><td>I = Scheduled Intraday Auction</td></tr>
<tr><td>UnscheduledAuction</td><td>5</td><td>U = Unscheduled Auction</td></tr>
<tr><td>ContinuousTrading</td><td>6</td><td>2 = Continuous Trading</td></tr>
<tr><td>AtMarketCloseTrading</td><td>7</td><td>3 = At Market Close Trading</td></tr>
<tr><td>OutOfMainSessionTrading</td><td>8</td><td>4 = Out Of Main Session Trading</td></tr>
<tr><td>TradeReportingOnExchange</td><td>9</td><td>5 = Trade Reporting (On Exchange)</td></tr>
<tr><td>TradeReportingOffExchange</td><td>10</td><td>6 = Trade Reporting (Off Exchange)</td></tr>
<tr><td>TradeReportingSystematicInternaliser</td><td>11</td><td>7 = Trade Reporting (Systematic Internaliser)</td></tr>
</table></td></tr>
<tr><td>mmtTransationCategory</td><td>MmtTransationCategory</td><td>Enum</td><td>1</td><td>MMT Transaction Category<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>DarkTrade</td><td>1</td><td>D = Dark Trade</td></tr>
<tr><td>TradePriceImprovement</td><td>2</td><td>R = Trade that has received price improvement</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>PackageTrade</td><td>3</td><td>Z = Package Trade (excluding Exchange For Physicals)</td></tr>
<tr><td>ExchangeForPhysicalsTrade</td><td>4</td><td>Y = Exchange For Physicals Trade</td></tr>
<tr><td>None</td><td>5</td><td>- = None apply (a standard trade for the Market Mechanism and Trading Mode)</td></tr>
</table>
</td></tr>
<tr><td>mmtNegotitationIndicator</td><td>MmtNegotitationIndicator</td><td>Enum</td><td>1</td><td>MMT Negotitation Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NegotiatedTrade</td><td>1</td><td>N = Negotiated Trade</td></tr>
<tr><td>NegotiatedTradeLiquidFinancialInstruments</td><td>2</td><td>1 = Negotiated Trade in Liquid Financial Instruments</td></tr>
<tr><td>NegotiatedTradeIlliquidFinancialInstruments</td><td>3</td><td>2 = Negotiated Trade in Illiquid Financial Instruments</td></tr>
<tr><td>NegotiatedTradeOther</td><td>4</td><td>3 = Negotiated Trade subject to conditions other than the current market price</td></tr>
<tr><td>NoNegotiatedTrade</td><td>5</td><td>- = No Negotiated Trade</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table>
<tr><td>PreTradeTransparencyWaiverIlliquidInstrument</td><td>6</td><td>4 = Pre-Trade Transparency Waiver for Illiquid Instrument on an SI (for RTS 1 only)</td></tr>
<tr><td>PreTradeTransparencyWaiverStandardMarketSize</td><td>7</td><td>5 = Pre-Trade Transparency Waiver for above Standard Market Size on an SI (for RTS 1 only)</td></tr>
<tr><td>PreTradeTransparencyWaiversIlqdAndSize</td><td>8</td><td>6 = Pre-Trade Transparency Waivers of ILQD and SIZE (for RTS 1 only)</td></tr>
</table></td></tr>
<tr><td>mmtAgencyCrossTradeIndicator</td><td>MmtAgencyCrossTradeIndicator</td><td>Enum</td><td>1</td><td>MMT Agency Cross Trade Indicator<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>AgencyCrossTrade</td><td>1</td><td>X = Agency Cross Trade</td></tr>
<tr><td>NoAgencyCrossTrade</td><td>2</td><td>- = No Agency Cross Trade</td></tr>
</table></td></tr>
<tr><td>mmtModificationIndicator</td><td>MmtModificationIndicator</td><td>Enum</td><td>1</td><td>MMT Modification Indicator<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>TradeCancellation</td><td>1</td><td>C = Trade Cancellation</td></tr>
<tr><td>TradeAmendment</td><td>2</td><td>A = Trade Amendment</td></tr>
<tr><td>NewTrade</td><td>3</td><td>- = New Trade</td></tr>
</table></td></tr>
<tr><td>mmtBenchmarkReferencePriceIndicator</td><td>MmtBenchmarkReferencePriceIndicator</td><td>Enum</td><td>1</td><td>MMT Benchmark / Reference Price Indicator<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table>
<tr><td>BenchmarkTrade</td><td>1</td><td>B = Benchmark Trade</td></tr>
<tr><td>ReferencePriceTrade</td><td>2</td><td>S = Reference Price Trade</td></tr>
<tr><td>NoBenchmarkorReferencePriceTrade</td><td>3</td><td>- = No Benchmark or Reference Price Trade</td></tr>
</table></td></tr>
<tr><td>mmtSpecialDividendIndicator</td><td>MmtSpecialDividendIndicator</td><td>Enum</td><td>1</td><td>MMT Special Dividend Indicator<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>SpecialDividendTrade</td><td>1</td><td>E = Special Dividend Trade</td></tr>
<tr><td>NoSpecialDividendTrade</td><td>2</td><td>- = No Special Dividend Trade</td></tr>
</table></td></tr>
<tr><td>mmtOffBookAutomatedIndicator</td><td>MmtOffBookAutomatedIndicator</td><td>Enum</td><td>1</td><td>MMT Off Book Automated Indicator<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>UnspecifiedOrNotApply</td><td>1</td><td>'- = Unspecified or does not apply</td></tr>
<tr><td>OffBookNonAutomated</td><td>2</td><td>M = Off Book Non-Automated</td></tr>
<tr><td>OffBookAutomated</td><td>3</td><td>Q = Off Book Automated</td></tr>
</table></td></tr>
<tr><td>mmtOrdinaryTradeIndicator</td><td>MmtOrdinaryTradeIndicator</td><td>Enum</td><td>1</td><td>MMT Ordinary Trade Indicator<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>PlainVanillaTrade</td><td>1</td><td>P = Plain-Vanilla Trade</td></tr>
<tr><td>NonPriceFormingTrade</td><td>2</td><td>I = Non-Price Forming Trade (formerly defined as a Technical Trade)</td></tr>
<tr><td>TradeNotContributingToPriceDiscovery</td><td>3</td><td>J = Trade not contributing to the price discovery process (formerly defined as a Technical Trade)</td></tr>
<tr><td>PricePending</td><td>4</td><td>N = Price is currently not available but pending</td></tr>
</table></td></tr>
</table>

Name	Data type	Kind	Length	Description
mmtAlgorithmicIndicator	MmtAlgorithmicIndicator	Enum	1	MMT Algorithmic Indicator **Name** | **Value** | **Description** AlgorithmicTrade | 1 | H = Algorithmic Trade NoAlgorithmicTrade | 2 | - = No Algorithmic Trade
mmtPostTradeDeferralReason	MmtPostTradeDeferralReason	Enum	1	MMT Post-Trade Deferral Reason **Name** | **Value** | **Description** ImmediatePublication | 1 | - = Immediate Publication NonImmediatePublication | 2 | 1 = Non-Immediate Publication NonImmediatePublicationLargeInScale | 3 | 2 = Non-Immediate Publication : Deferral for 'Large in Scale' NonImmediatePublicationIlliquidInstrument | 4 | 3 = Non-Immediate Publication : Deferral for 'Illiquid Instrument' (for RTS 2 only) NonImmediatePublicationSizeSpecific | 5 | 4 = Non-Immediate Publication : Deferral for 'Size Specific' (for RTS 2 only) NonImmediatePublicationIlqdAndSize | 6 | 5 = Non-Immediate Publication : Deferrals of ILQD and SIZE (for RTS 2 use only) NonImmediatePublicationIlqdAndLrgs | 7 | 6 = Non-Immediate

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td><table><tr><td></td><td></td><td>Publication : Deferrals of ILQD and LRGS (for RTS 2 use only)</td></tr></table></td></tr>
<tr><td>mmtPostTradeDeferralType</td><td>MmtPostTradeDeferralType</td><td>Enum</td><td>1</td><td>MMT Post-Trade Deferral Type
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>LimitedDetailsTrade</td><td>1</td><td>1 = Limited Details Trade</td></tr>
<tr><td>DailyAggregatedTrade</td><td>2</td><td>2 = Daily Aggregated Trade</td></tr>
<tr><td>VolumeOmissionTrade</td><td>3</td><td>3 = Volume Omission Trade</td></tr>
<tr><td>FourWeeksAggregationTrade</td><td>4</td><td>4 = Four Weeks Aggregation Trade</td></tr>
<tr><td>IndefiniteAggregationTrade</td><td>5</td><td>5 = Indefinite Aggregation Trade</td></tr>
<tr><td>VolumeOmissionTradeAggregated</td><td>6</td><td>6 = Volume Omission Trade, eligible for subsequent enrichment in aggregated form</td></tr>
<tr><td>NotApplicable</td><td>7</td><td>- = Not applicable / No relevant deferral or enrichment type</td></tr>
</table></td></tr>
</table>

11.2.40.TRADECOLLARS:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of financial instrument.</td></tr>
<tr><td>collarType</td><td>CollarType</td><td>Enum</td><td>1</td><td>Collar type.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Static</td><td>1</td><td></td></tr>
<tr><td>Dynamic</td><td>2</td><td></td></tr>
</table></td></tr>
</table>

Name	Data type	Kind	Length	Description
price	Price	Alias (Number)	8	The reference price.
lower	Bound	Alias (Number)	8	Value of the lower Price collar.
upper	Bound	Alias (Number)	8	Value of the upper Price collar.

11.2.41. TRADINGPHASESCHEDULEENTRY:

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>header</td><td>Header</td><td>Struct</td><td>39</td><td>Message header.</td></tr>
<tr><td>tradingScheduleId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Trading schedule ID.</td></tr>
<tr><td>tradingPhaseId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Trading phase ID.</td></tr>
<tr><td>tradingPhaseSettlementCycle</td><td>u8</td><td>Primitive</td><td>1</td><td>Settlement cycle for transactions made during a continuous phase.</td></tr>
<tr><td>tradingPhaseMaxBlockSettlementCycle</td><td>u8</td><td>Primitive</td><td>1</td><td>Max settlement cycle for block transactions.</td></tr>
<tr><td>staticCollarVolatilityAuctionId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Static collar volatility auction ID.</td></tr>
<tr><td>dynamicCollarVolatilityAuctionId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Dynamic collar volatility auction ID.</td></tr>
<tr><td>tradingPhaseStartTime</td><td>Timestamp</td><td>Alias (u64)</td><td>8</td><td>Trading phase start time.</td></tr>
<tr><td>tradingPhaseType</td><td>TradingPhaseType</td><td>Enum</td><td>1</td><td>Type of matching algorithm.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>ContinuousPriceTime</td><td>1</td><td>Continuous price and time.</td></tr>
<tr><td>ContinuousRefPriceTime</td><td>2</td><td>Continuous time at reference price.</td></tr>
<tr><td>ContinuousLastAuctionTime</td><td>3</td><td>Trade at last.</td></tr>
<tr><td>AuctionOpening</td><td>4</td><td>Auction opening.</td></tr>
<tr><td>BlockExactMatching</td><td>5</td><td>Block transaction, exact matching.</td></tr>
<tr><td>NoTradingClosed</td><td>6</td><td>Trading disabled, no matching.</td></tr>
<tr><td>Uncrossing</td><td>7</td><td>Uncrossing.</td></tr>
<tr><td>AuctionClosing</td><td>8</td><td>Auction closing.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>AuctionIntraday</td><td>9</td><td>Auction intraday.</td></tr>
<tr><td>AuctionVolatilityStatic</td><td>10</td><td>Auction volatility static.</td></tr>
<tr><td>AuctionVolatilityDynamic</td><td>11</td><td>Auction volatility dynamic.</td></tr>
<tr><td>AuctionExtendedVolatilityStatic</td><td>12</td><td>Auction extended volatility static.</td></tr>
<tr><td>AuctionExtendedVolatilityDynamic</td><td>13</td><td>Auction extended volatility dynamic.</td></tr>
<tr><td>NoTradingEarlyMonitoring</td><td>14</td><td>Early Monitoring</td></tr>
<tr><td>NoTradingLateMonitoring</td><td>15</td><td>Late Monitoring</td></tr>
<tr><td>Hybrid</td><td>16</td><td>Hybrid normal trading</td></tr>
<tr><td>HybridBuyOnly</td><td>17</td><td>Hybrid buy only phase</td></tr>
<tr><td>HybridPreTrade</td><td>18</td><td>Hybrid pre trade phase</td></tr>
<tr><td>UnsuspensionAuction</td><td>19</td><td>Auction after suspension</td></tr>
<tr><td>Ipo</td><td>20</td><td>Initial public offering phase</td></tr>
</table>
</td></tr>
<tr><td>auctionType</td><td>AuctionType</td><td>Enum</td><td>1</td><td>Type of auction.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>NotApplicable.</td></tr>
<tr><td>OpeningAuction</td><td>2</td><td>Opening auction.</td></tr>
<tr><td>ClosingAuction</td><td>3</td><td>Closing auction.</td></tr>
<tr><td>IntradayAuction</td><td>4</td><td>Intraday auction.</td></tr>
<tr><td>VolatilityAuctionStatic</td><td>5</td><td>Volatility auction after static collars breach.</td></tr>
<tr><td>VolatilityAuctionDynamic</td><td>6</td><td>Volatility auction after dynamic collars breach.</td></tr>
<tr><td>ExtendedVolatilityAuctionStatic</td><td>7</td><td></td></tr>
<tr><td>ExtendedVolatilityAuctionDynamic</td><td>8</td><td></td></tr>
<tr><td>UnsuspensionAuction</td><td>9</td><td></td></tr>
</table>
</td></tr>
<tr><td>uncrossing</td><td>bool</td><td>Primitive</td><td>1</td><td>True if the phase includes an uncrossing.</td></tr>
<tr><td>extStaticCollarVolatilityAuctionId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of Static Collar Volatility Auction.</td></tr>
<tr><td>extDynamicCollarVolatilityAuctionId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of Dynamic Collar Volatility Auction.</td></tr>
<tr><td>lastAuctionPhaseId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of last auction phase.</td></tr>
</table>

11.2.42. WEEKPLAN:

Name	Data type	Kind	Length	Description
header	Header	Struct	39	Message header.
calendarId	ElementId	Alias (u32)	4	Calendar ID.
weekdays	Weekdays	Array (bool)	7	Boolean array of weekdays (Monday to Sunday) open or closed for trading; true – weekday open for trading, false – weekday closed for trading.